SAS Viya Workshop

Build Your Analytics Skills with SAS Viya: Data Visualization, Data Preparation, and Machine Learning

12 November 2026

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https://github.com/pvateekul/sas-viya-2026/

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About me

- Professor at Department of Computer Engneering, Faculty of Engineering, Chulalongkorn University
  - Specialized in applying AI/ML/DL into various domains & design Big Data platform (DB/DW)
  - Course: Data Science and Data Engineering, Data Warehousing, Natural Language Processing, etc.
- Project advisors for many companies, specialized in intelligent customer data platforms (data pipeline, model construction/deployment, ML operations)
- Certified SAS instructor
- Certified instructor for Deep Learning Institute
- Parts of NVAITC (NVIDIA AI Technology Center)



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About me: SAS Experience

Be an instructor for SAS Thailand for about 14 years

SAS Programming (PG1, PG2, PG3)

SAS Macro Programming

SAS Enterprise Miner (EM) + Credit Scoring

SAS Data Integration (DI)

SAS Viya (VA1, VA2, VS, VDML, CPML, BDID, VSTUC)

SAS Model Studio

SAS Data Studio

Etc. (SAS for Hadoop, SAS for R)

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Outline

1. Overview of the Analytics Journey
2. SAS Viya Setup and Solution Tour

3. EDA with SAS Visual Analytics
4. Data Engineering and Preparation with SAS
5. Machine Learning with SAS
6. Data Governance with SAS
7. Conclusion

Schedule

- 9:00AM - 9:30AM: Intro
- 9:30AM - 10:45AM
- 15 mins break
- 11:00AM - 12PM
- 1PM - 2:30PM
- 15 mins break
- 2:45PM - 3:30PM
- 3:30PM - 4PM: QA



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1. Overview of the Analytics Journey

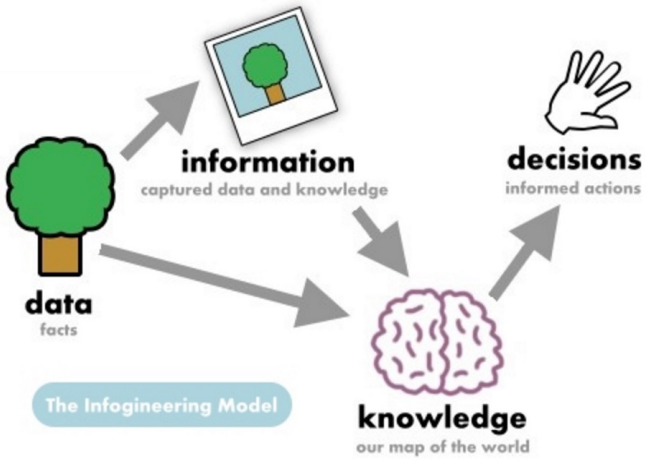
Introduction to SAS Viya

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sas
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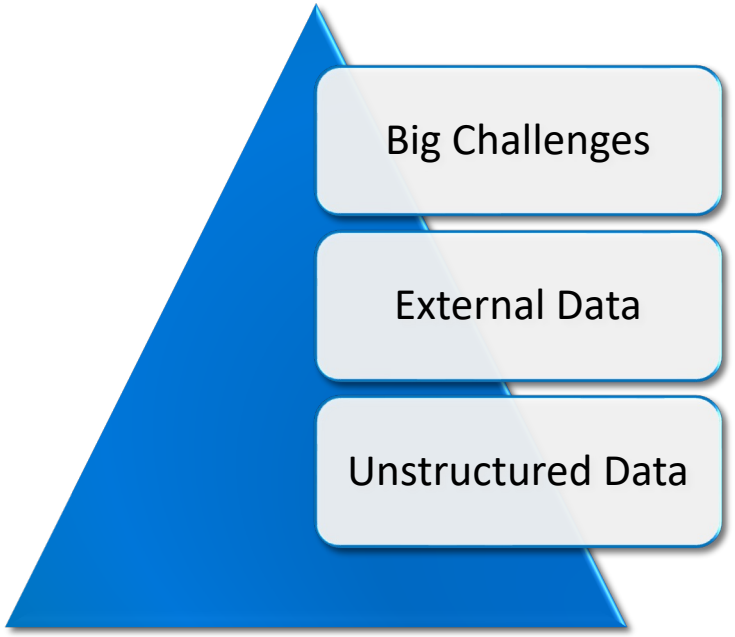
Big Data Analytics

- It is a process of examining Big Data to uncover useful information and knowledge.
- More data means better decision!

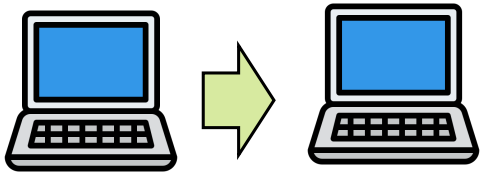








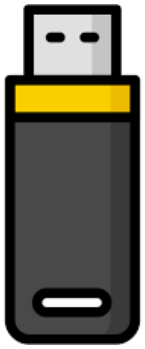
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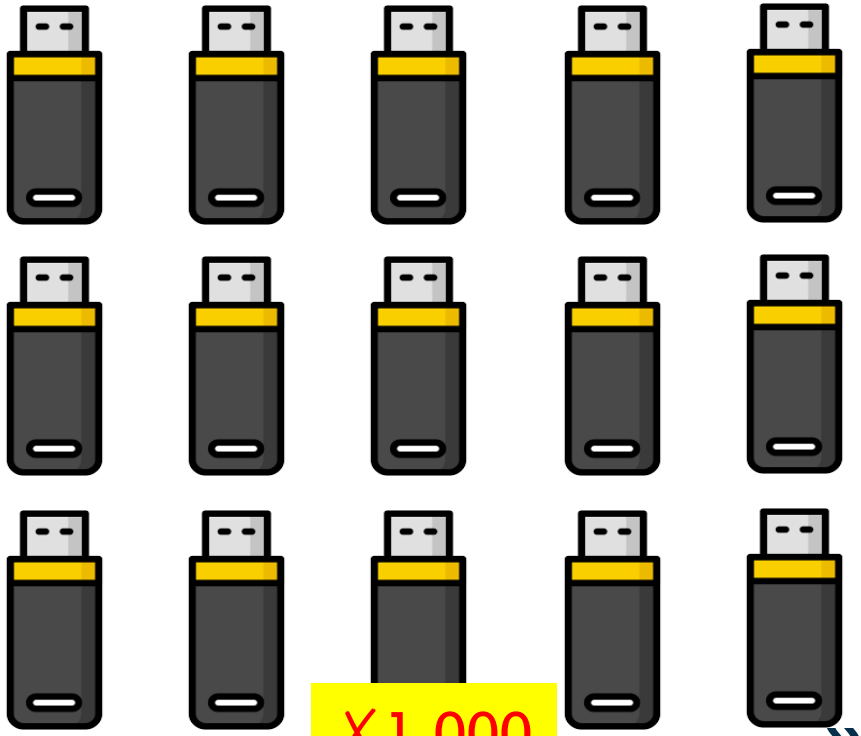
Big Data Challenges

Same tasks, but much more difficult!

2MB



2TB



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Big Data Solution





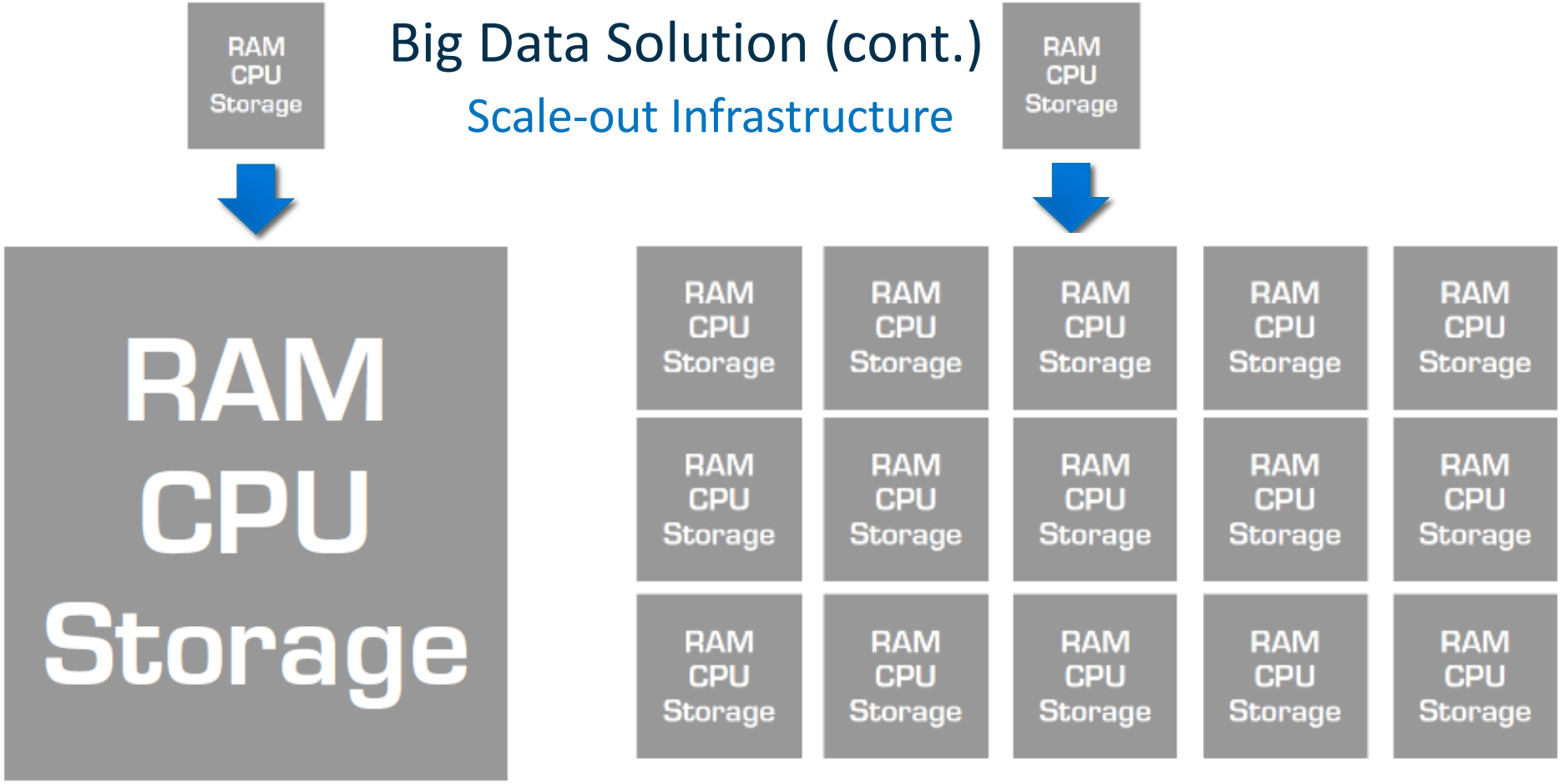
INFRASTRUCTURE

ALGORITHM

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Big Data Solution (cont.)

Scale-out Infrastructure



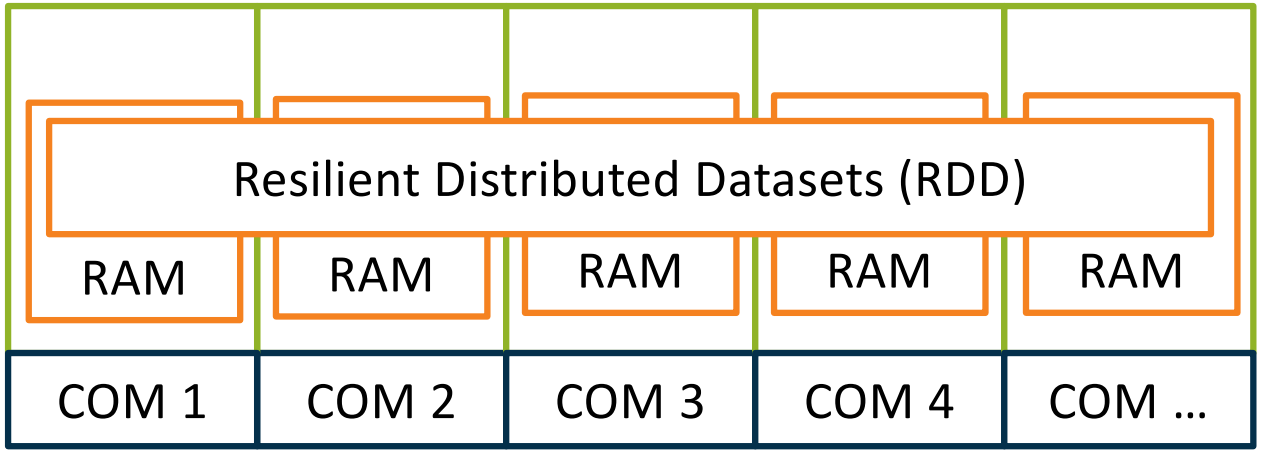
Vertical Scaling
(Scale-up)

Horizontal Scaling
(Scale-out)

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Big Data Solution (cont.)

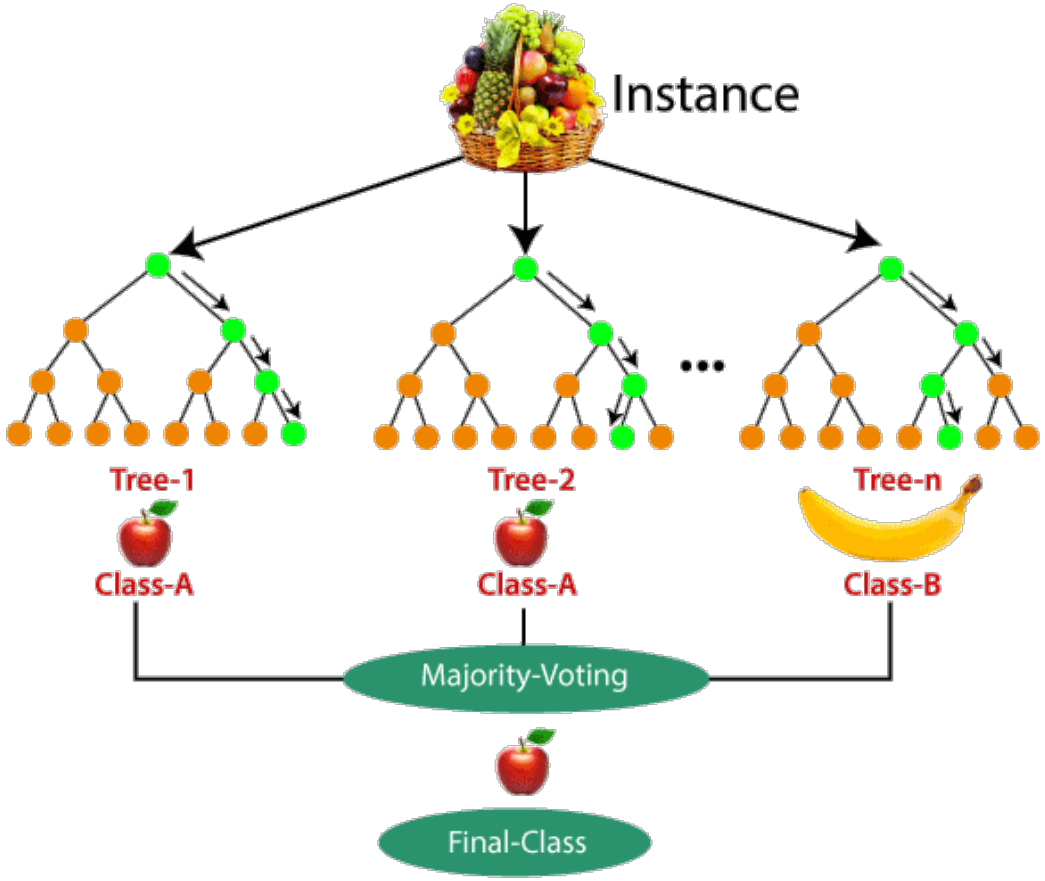
In-memory & Distributed Computing



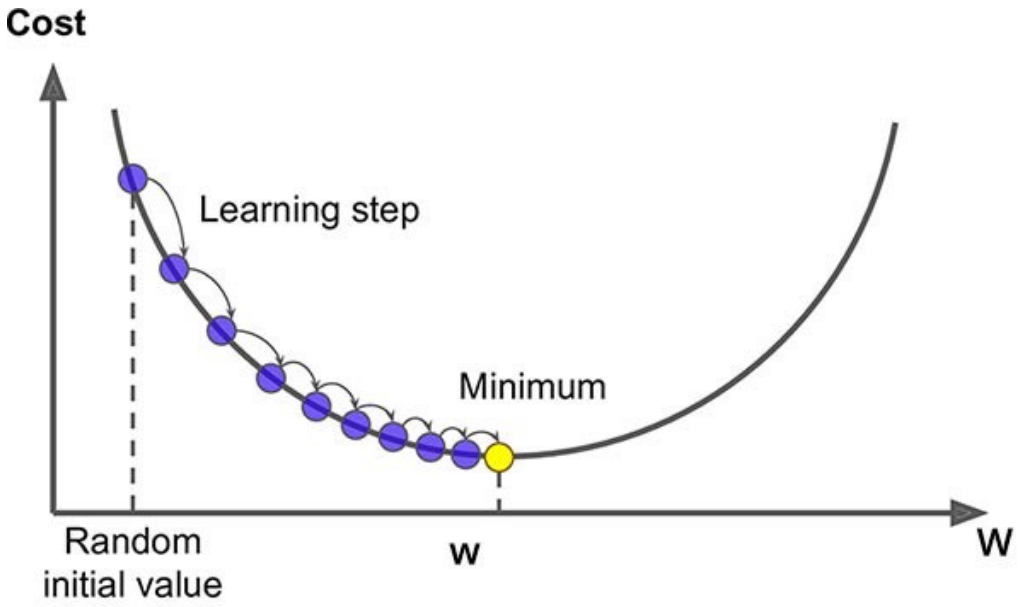
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Big Data Solution (cont.)

Optimize model by parallelizing jobs and reducing iterations



Random Forest



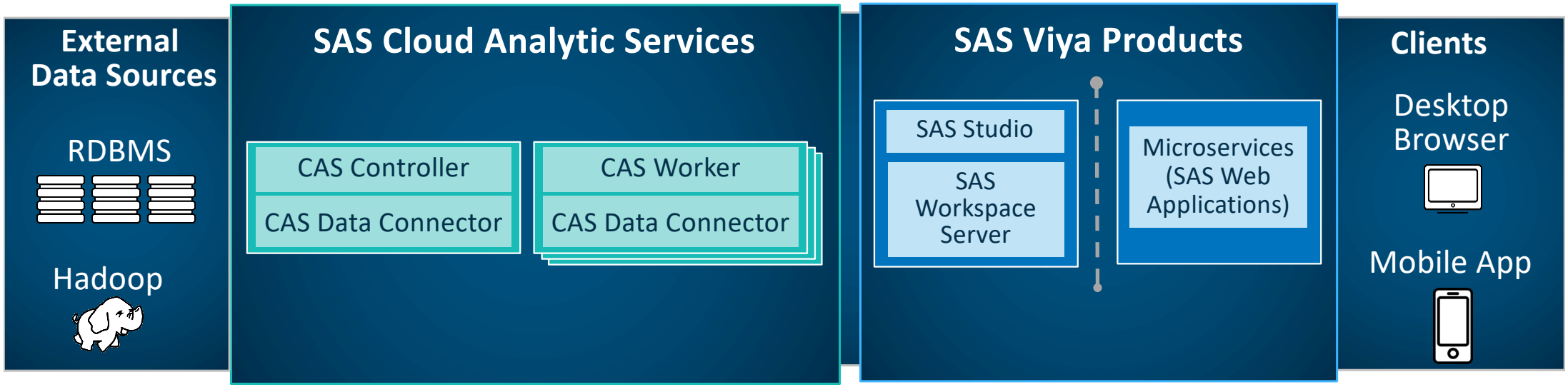
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SAS Viya

Big Data Analytics & Cloud Solution

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SAS Viya Architecture



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SAS Cloud Analytic Services (CAS)



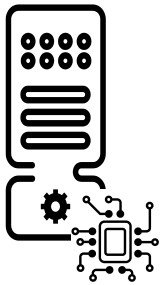
Cloud Analytic Services (CAS) is the server that provides the run-time environment for data management and analytics with SAS Viya.

CAS uses a high-performance, in-memory engine and a distributed architecture to execute multi-threaded analytic code. It can rapidly process requests against data of any size. CAS provides the following features:

- speed
- resiliency
- scalability
- node-to-node communication
- worker node fault tolerance

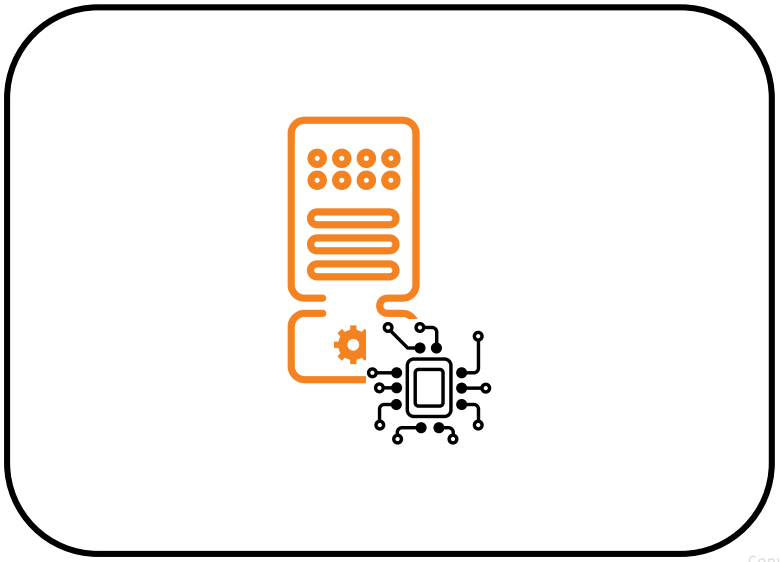
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SAS Viya Architecture

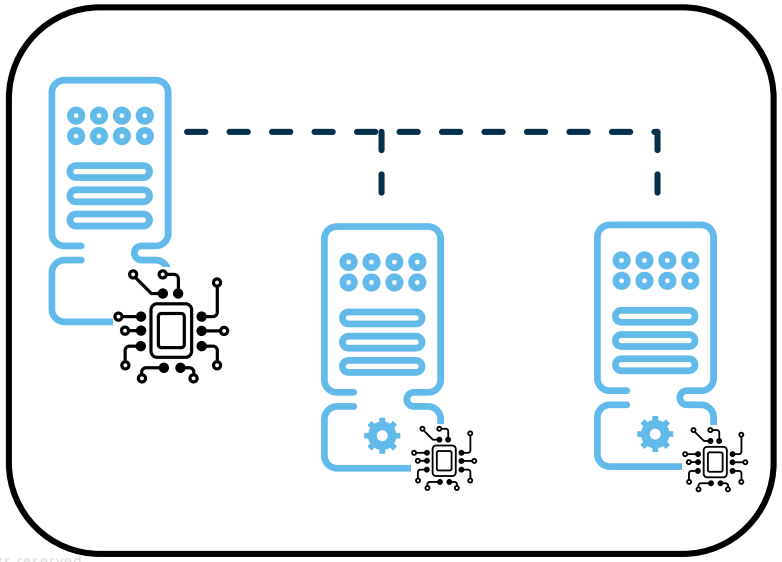


SAS Cloud Analytic Services (CAS)

Symmetric Multiprocessing (SMP)

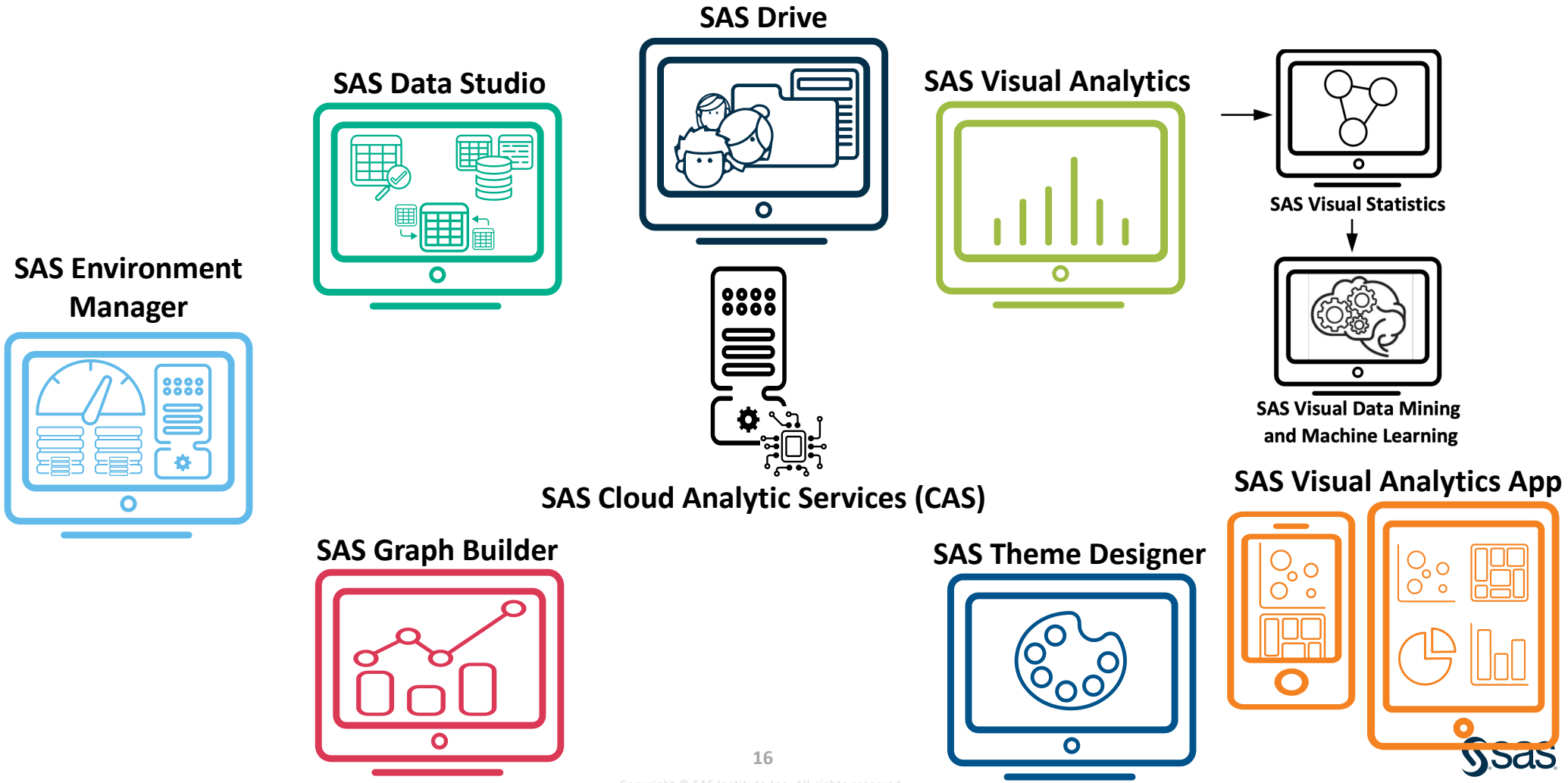


Massively Parallel Processing (MPP)



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SAS Viya Applications



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Base SAS → SAS Viya (Cloud)

Base SAS	SAS Viya	Note
Compute Server	CAS Servers	Infrastructure
SAS Enterprise Guide (EG)	SAS Studio	Coding, statistical analysis
SAS Enterprise Miner (EM)	SAS Model Studio	Machine Learning (full ML pipeline)
	SAS Visual Analytics (VA)	BI Dashboard (PowerBI)
	SAS Visual Statistics (VS)	Quick ML model
SAS Data Integration (DI)	SAS Data Studio	ETL
	SAS Intelligent Decisioning (ID)	Action (campaign) automation
	SAS Data Governance	Data Governance
	SAS Environment Manager	Admin, job scheduling

The Analytics Journey Overview

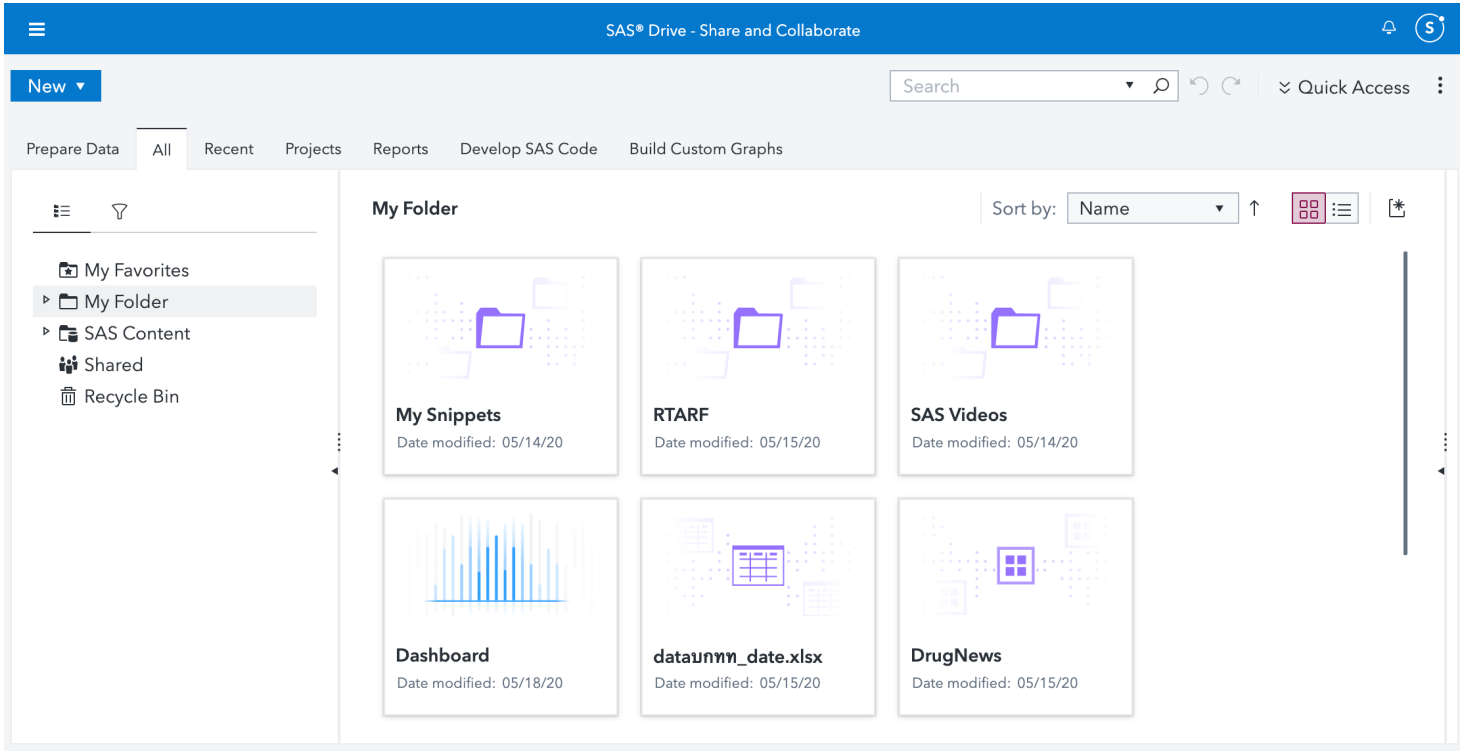


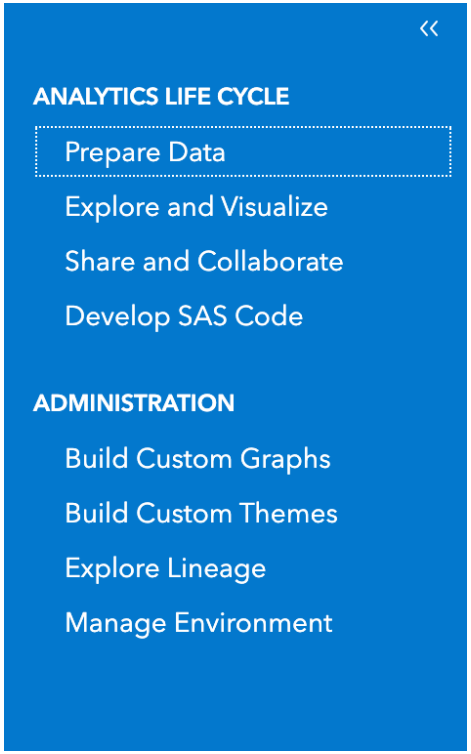
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SAS Drive (Home Page)

SAS Data Explorer



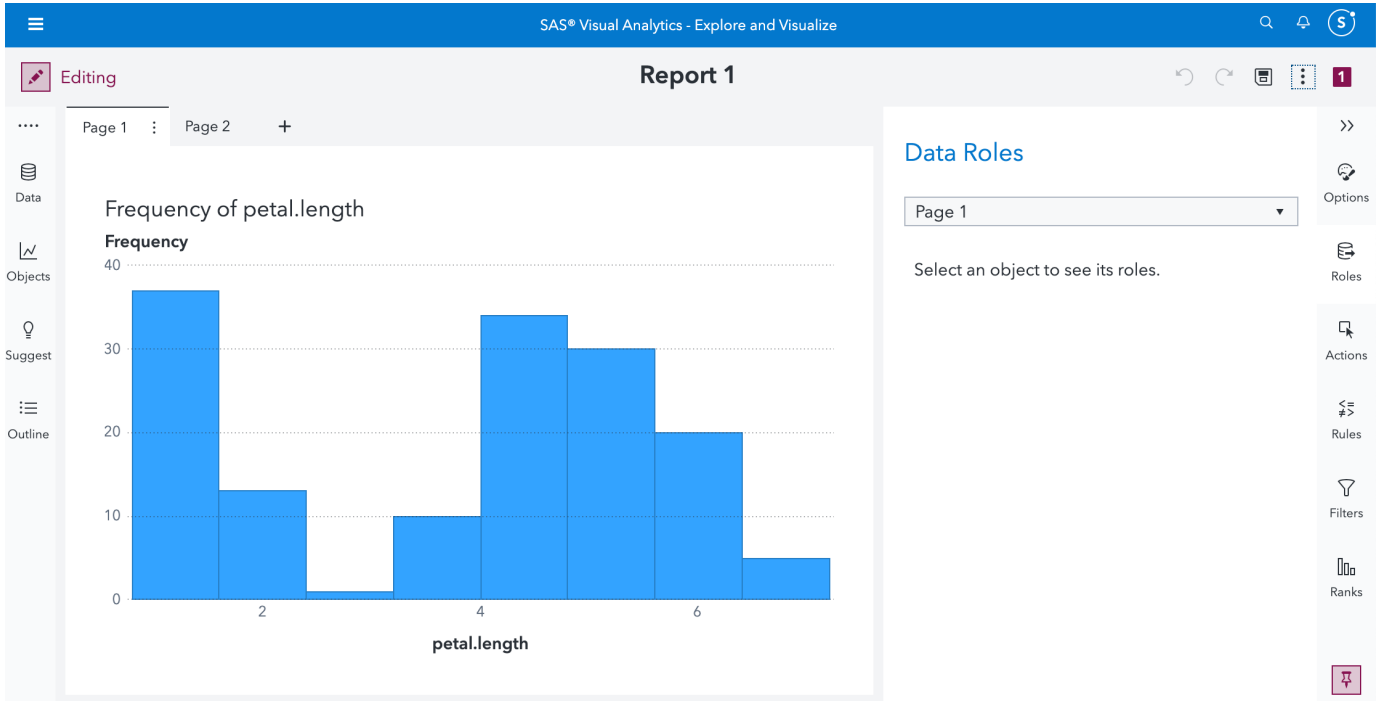


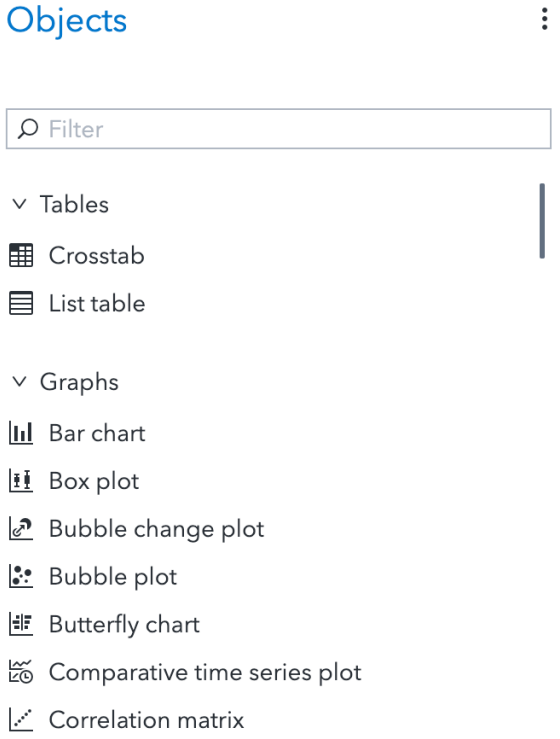


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SAS Visual Analytics (VA): Dashboard



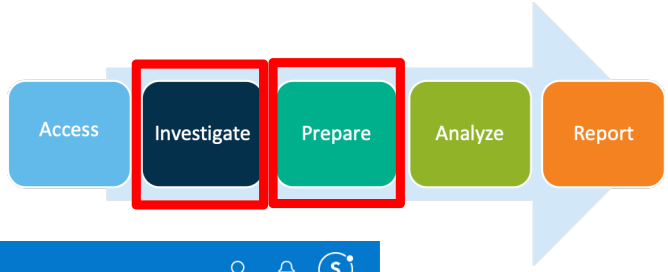


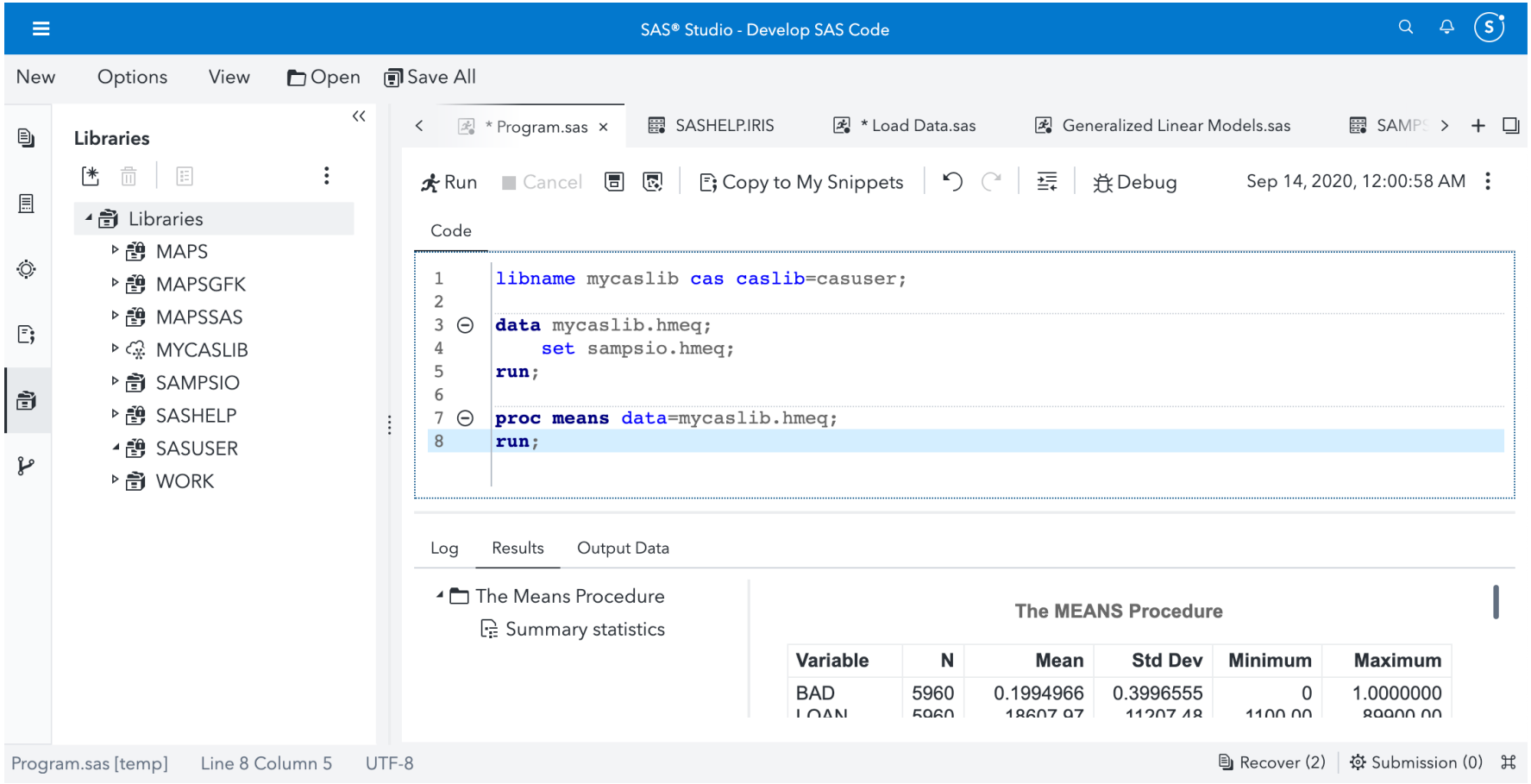


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SAS Studio - Develop SAS Code

SAS Studio Flow

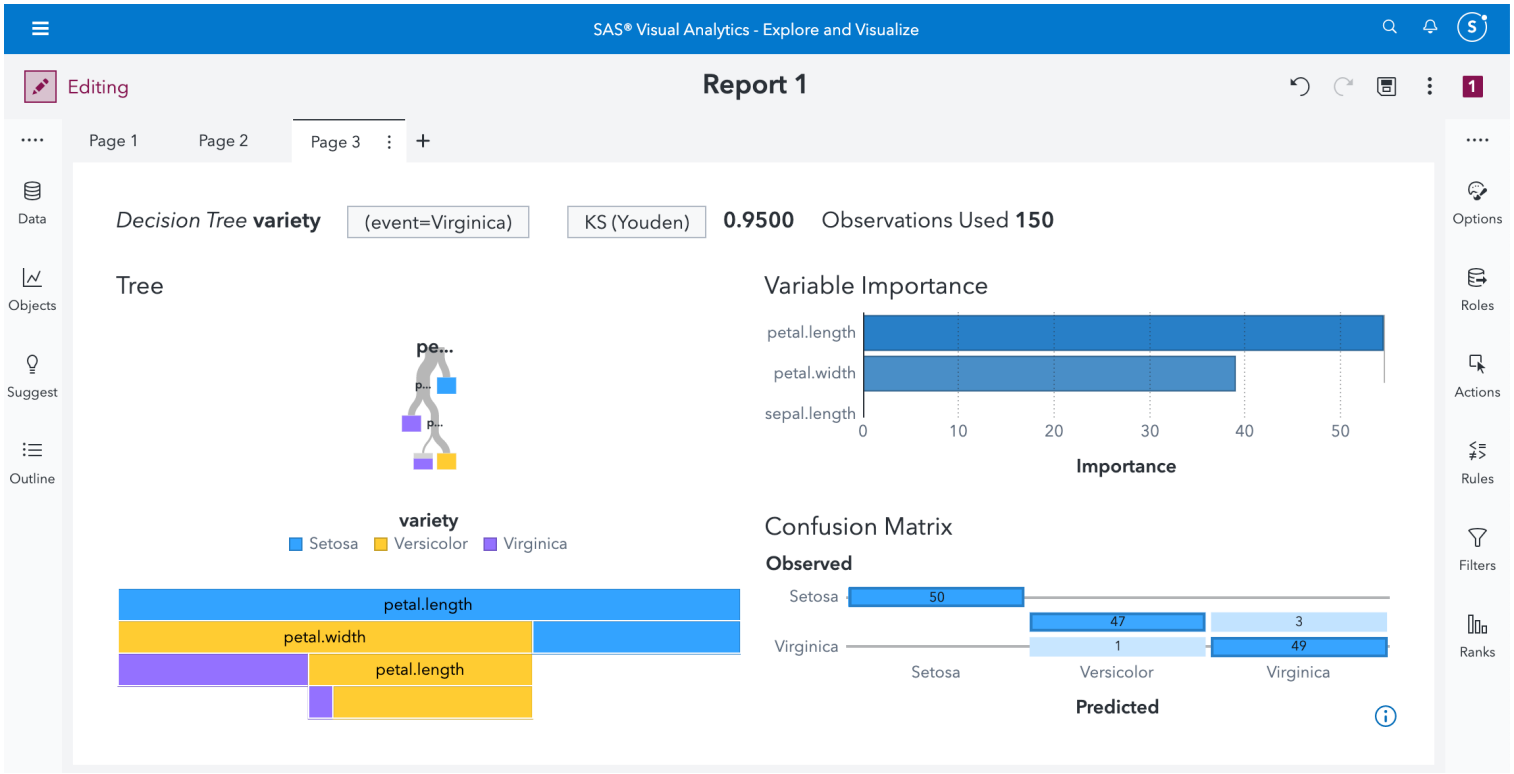


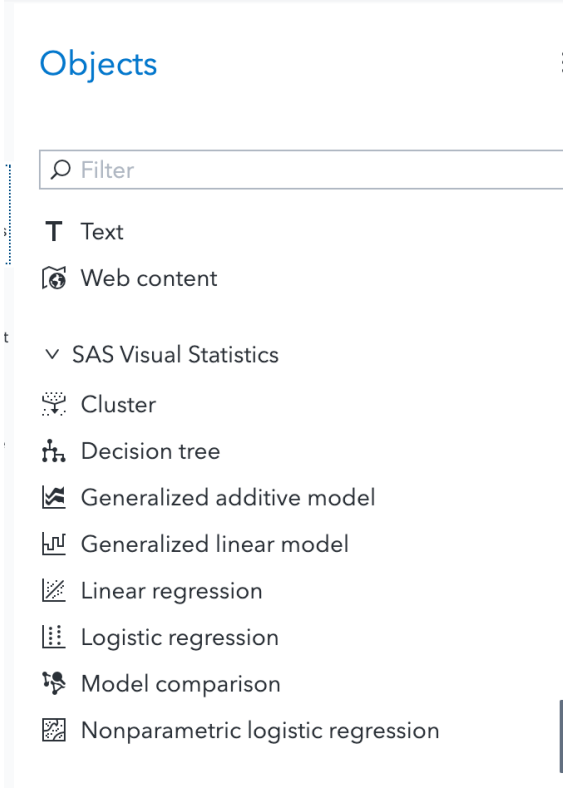


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SAS Visual Statistics (VS): ML



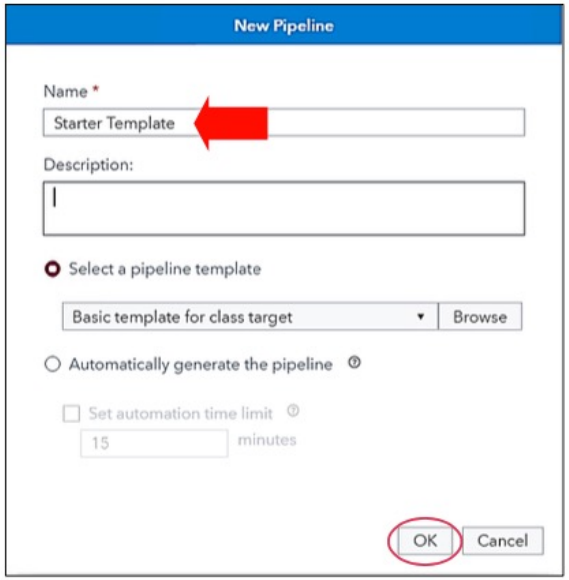


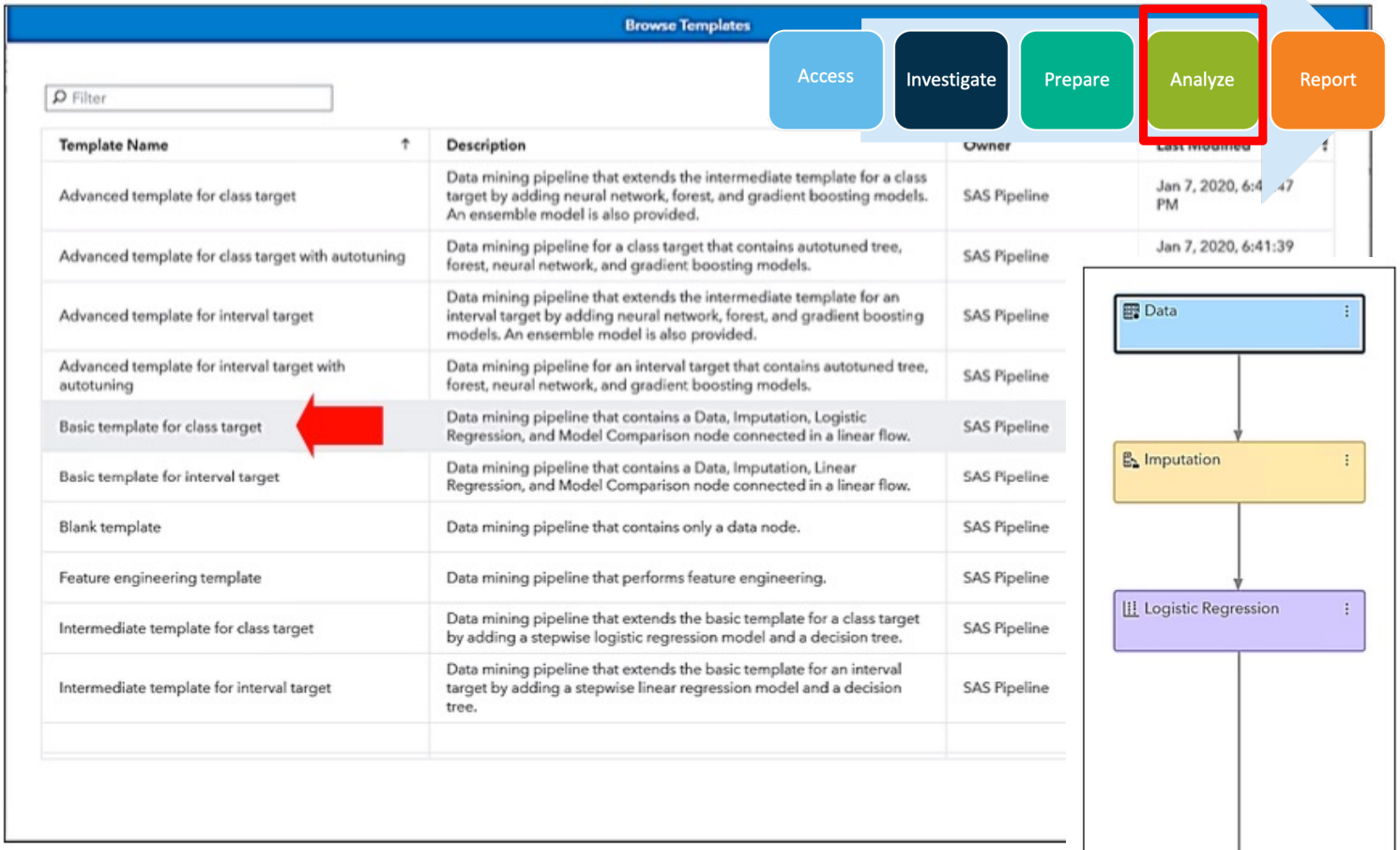


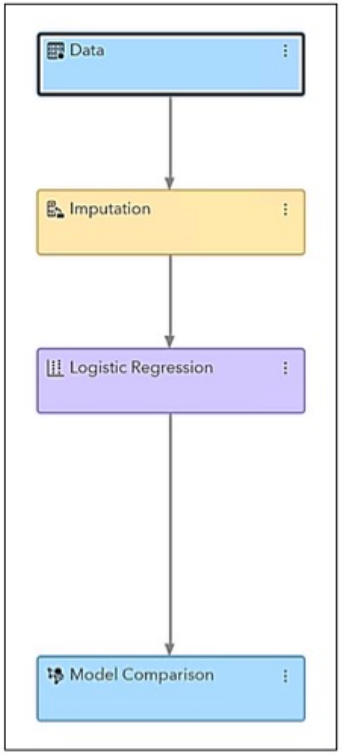
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SAS Model Studio - Develop Model Pipeline









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2. SAS Viya Setup and Solution Tour

Lab only & no slides

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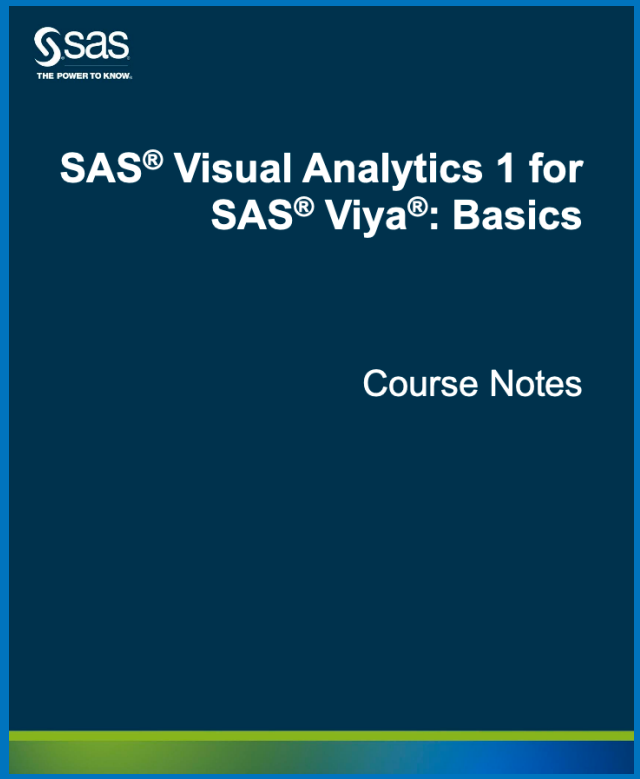


3) Visual Analytics (VA): EDA with SAS Visual Analytics

2.1) Exploring data with charts & graphs

2.2) Creating interactive reports

2.3) Working with display rules



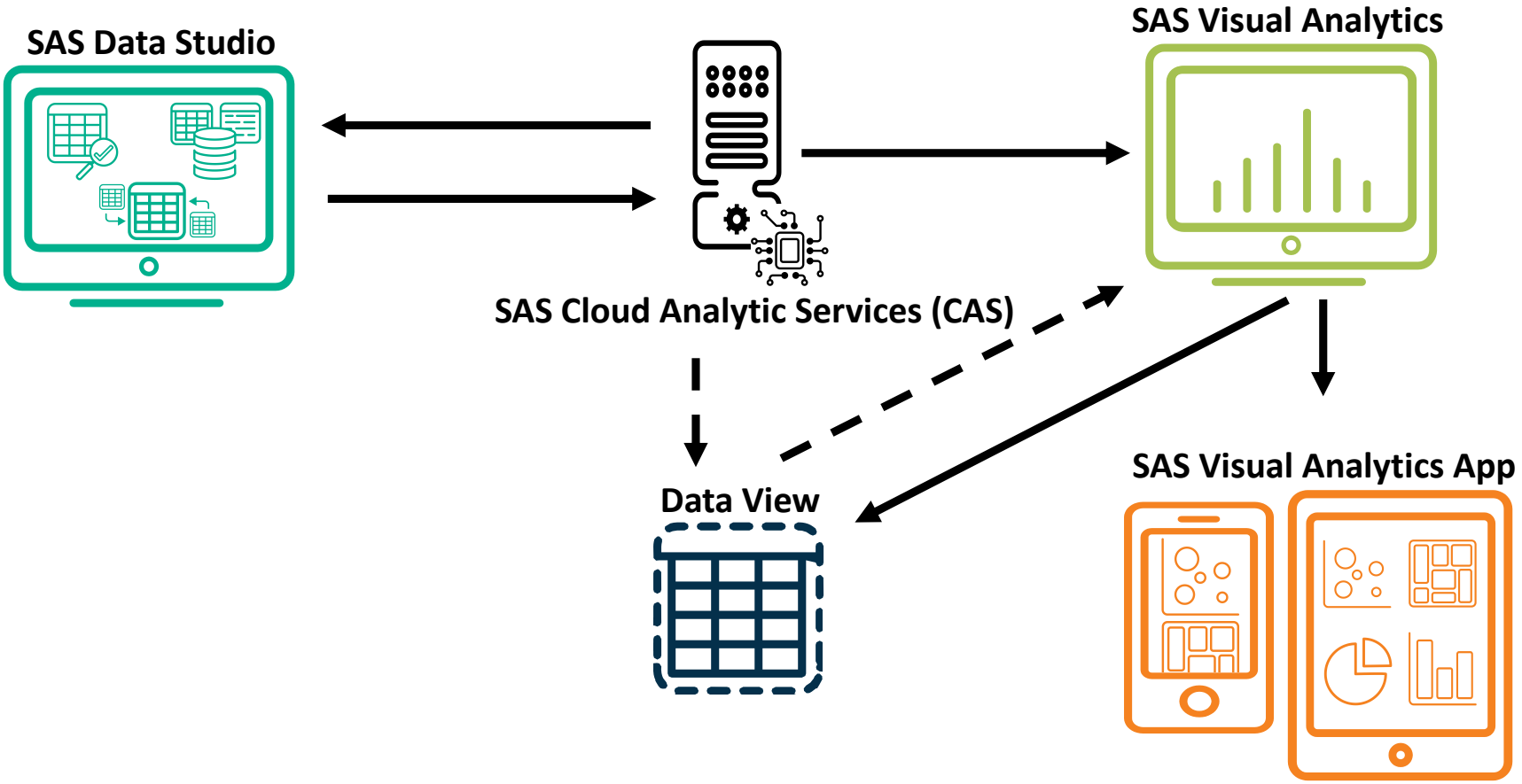
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2.1) Exploring data with charts & graphs

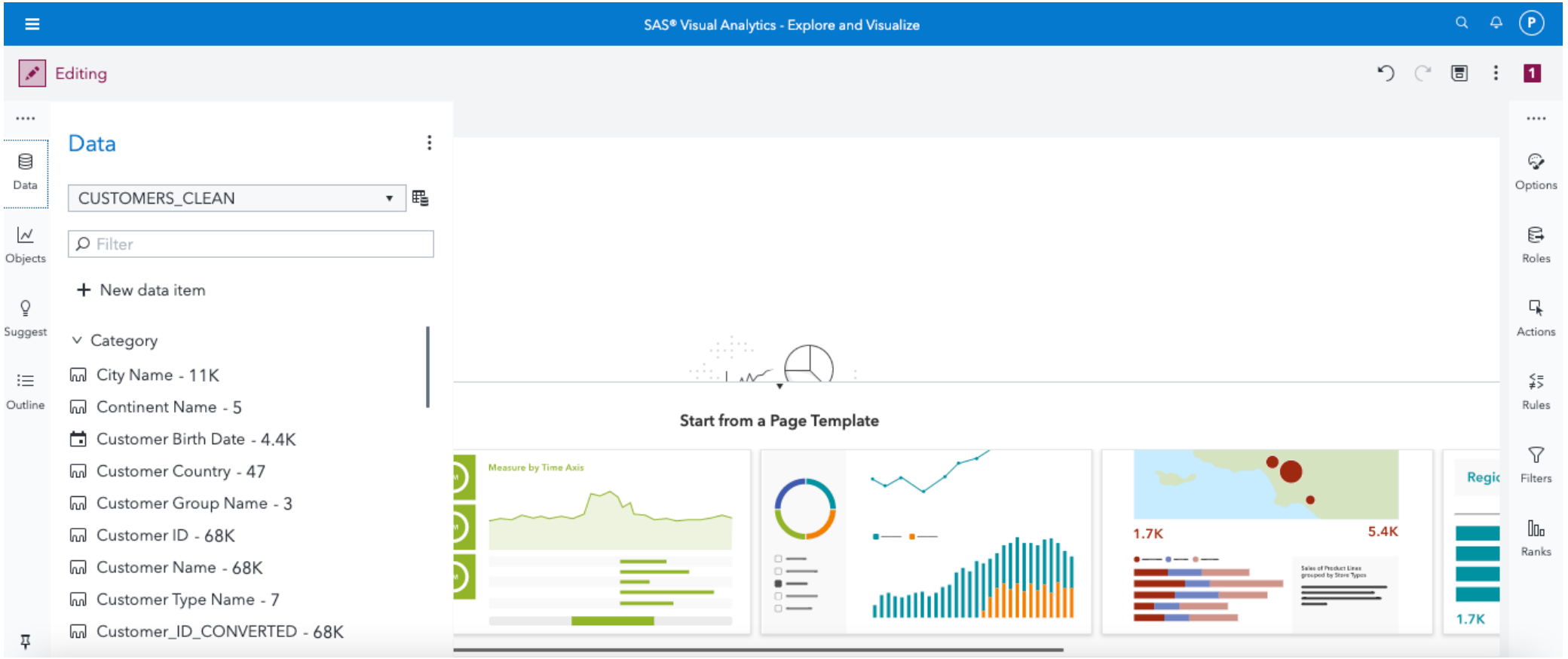
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SAS Data Studio versus Visual Analytics



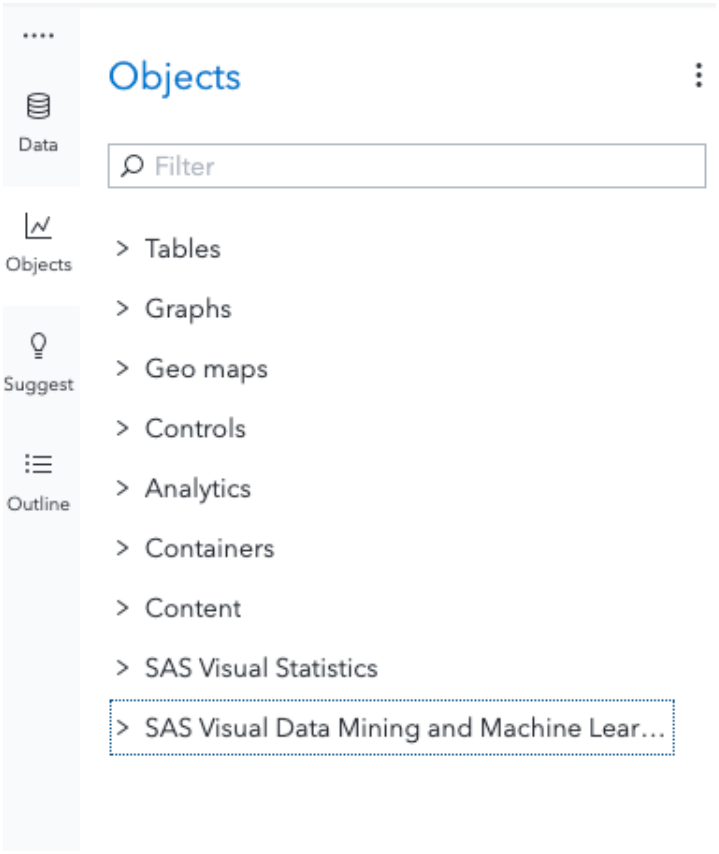
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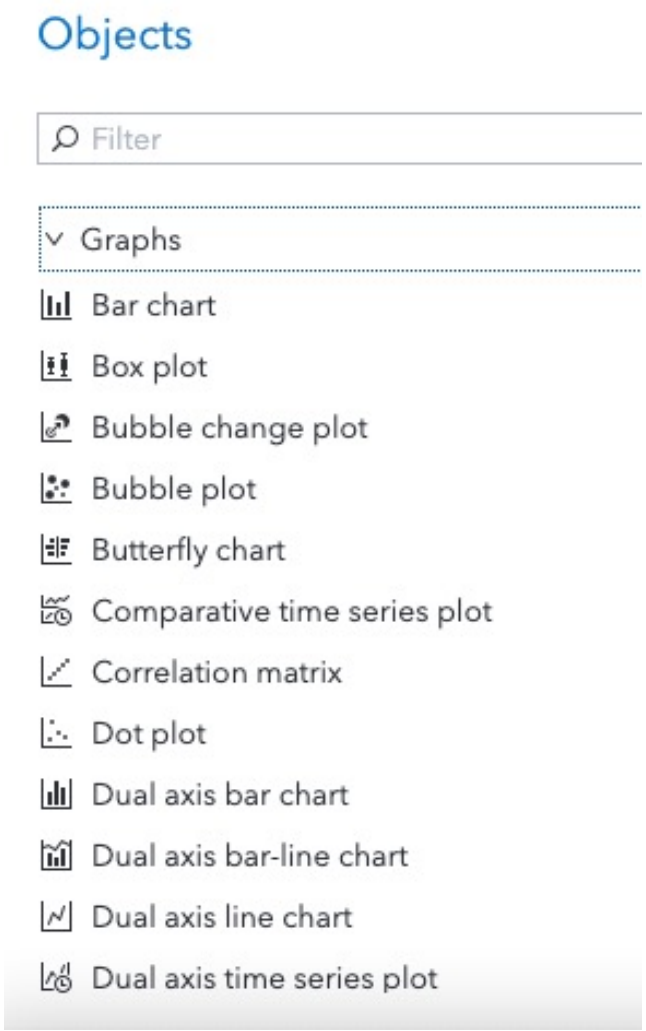
Prepare Data in SAS VA

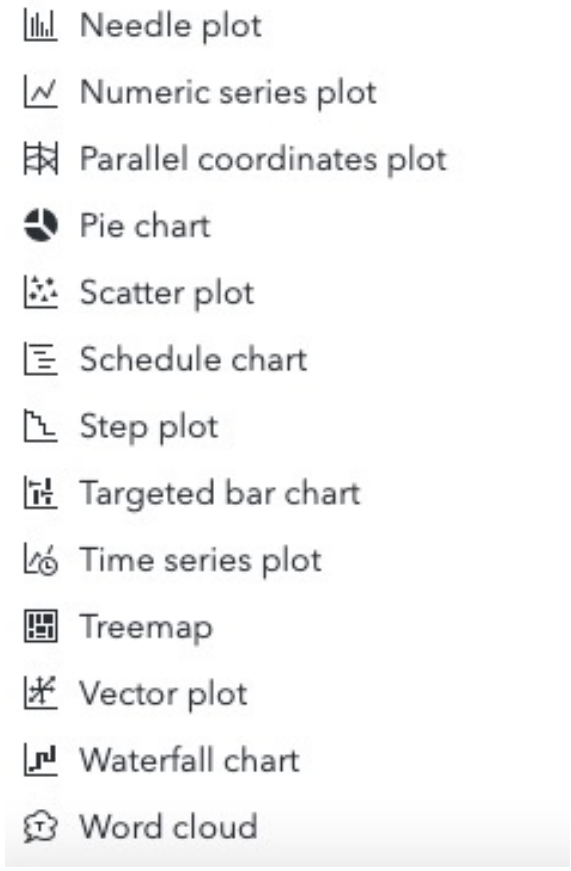


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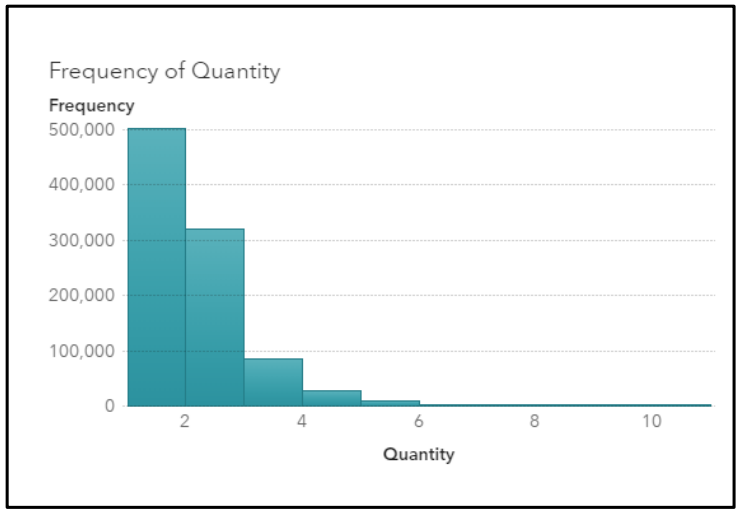
Objects in SAS VA







Objects: Graphs (Descriptive)



Use a *histogram* to view the distribution of a single measure.

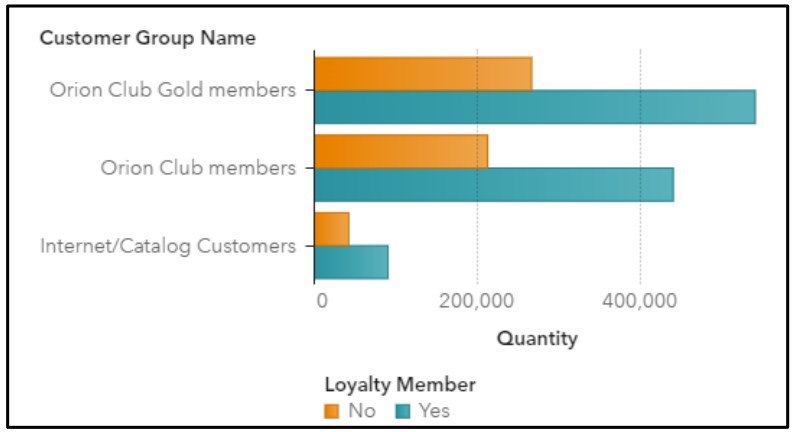


Use a *box plot* to view information about the variability of the data and extreme values.

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Objects: Graphs (Descriptive)

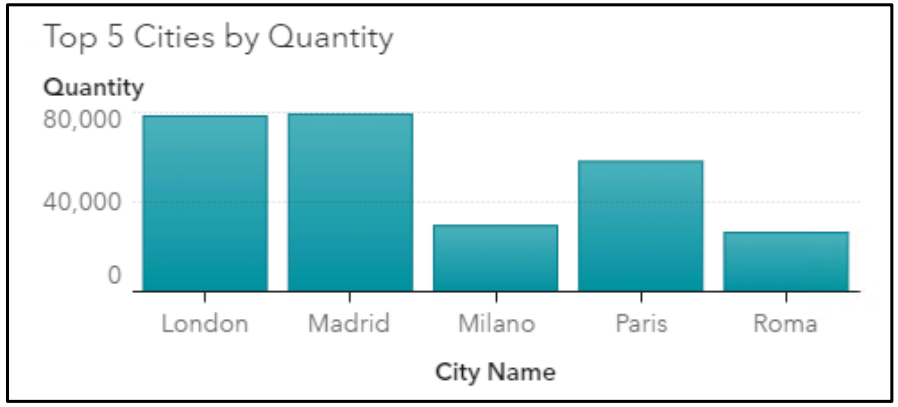
Nominal values



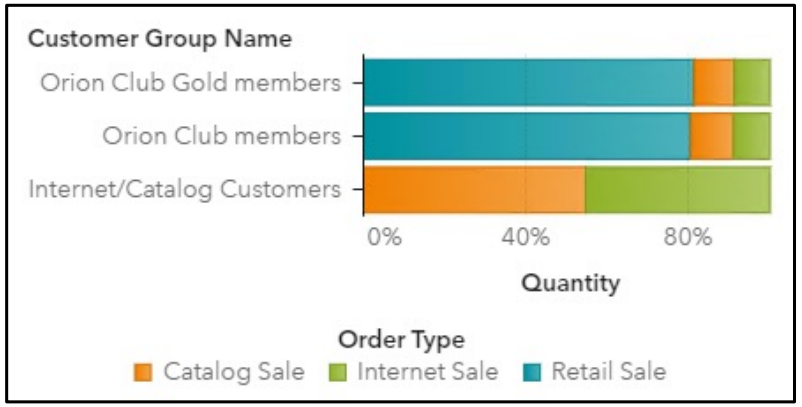
Time series data



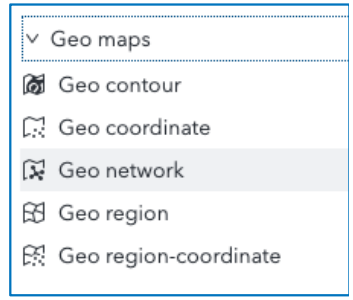
Rankings



Parts of a whole



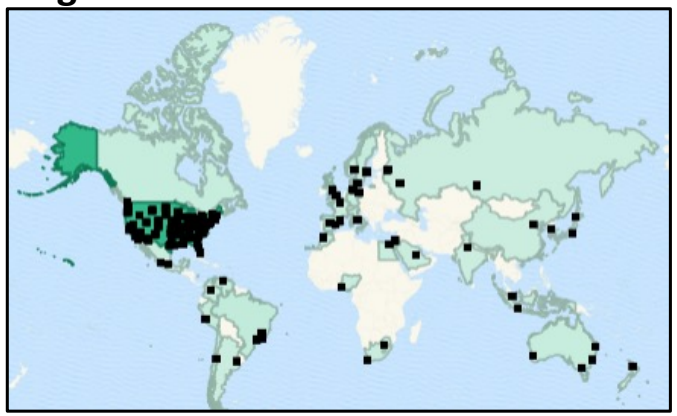
Objects: Graphs (Geography)



Coordinate



Region-Coordinate



Use a *region* or *region-coordinate map* only when there is an even distribution of values within each region.

Use a *geo contour map* to show very dense data.

Contour



Region

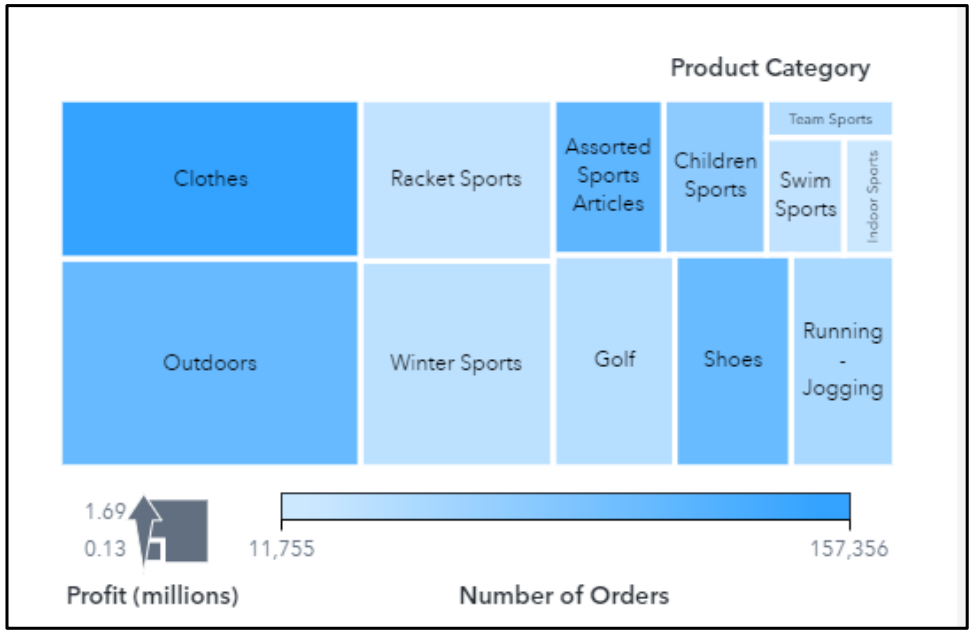


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Objects: Graphs (Analysis): X, Y, Color, Size



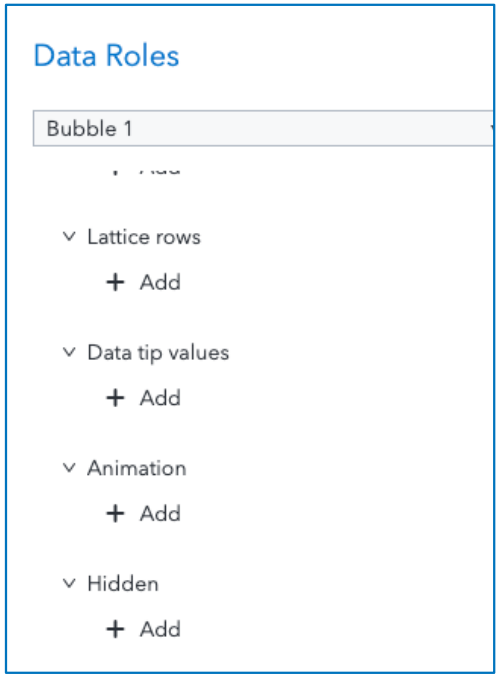
Use a *bubble plot* to display three dimensions of data (horizontal location, vertical location, size of bubble) for some group of category values.

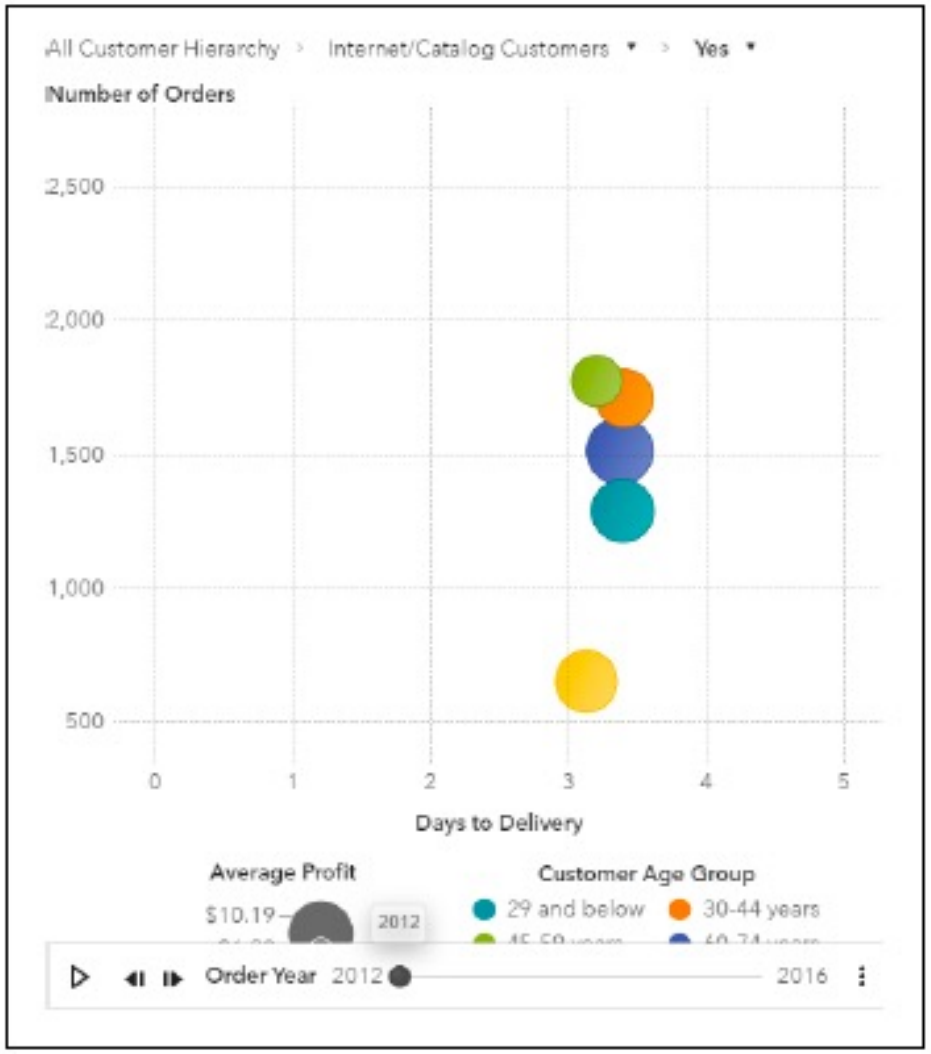


Use a *treemap* to display a lot of information in a small amount of space. Use size and color to draw attention to specific areas of interest.

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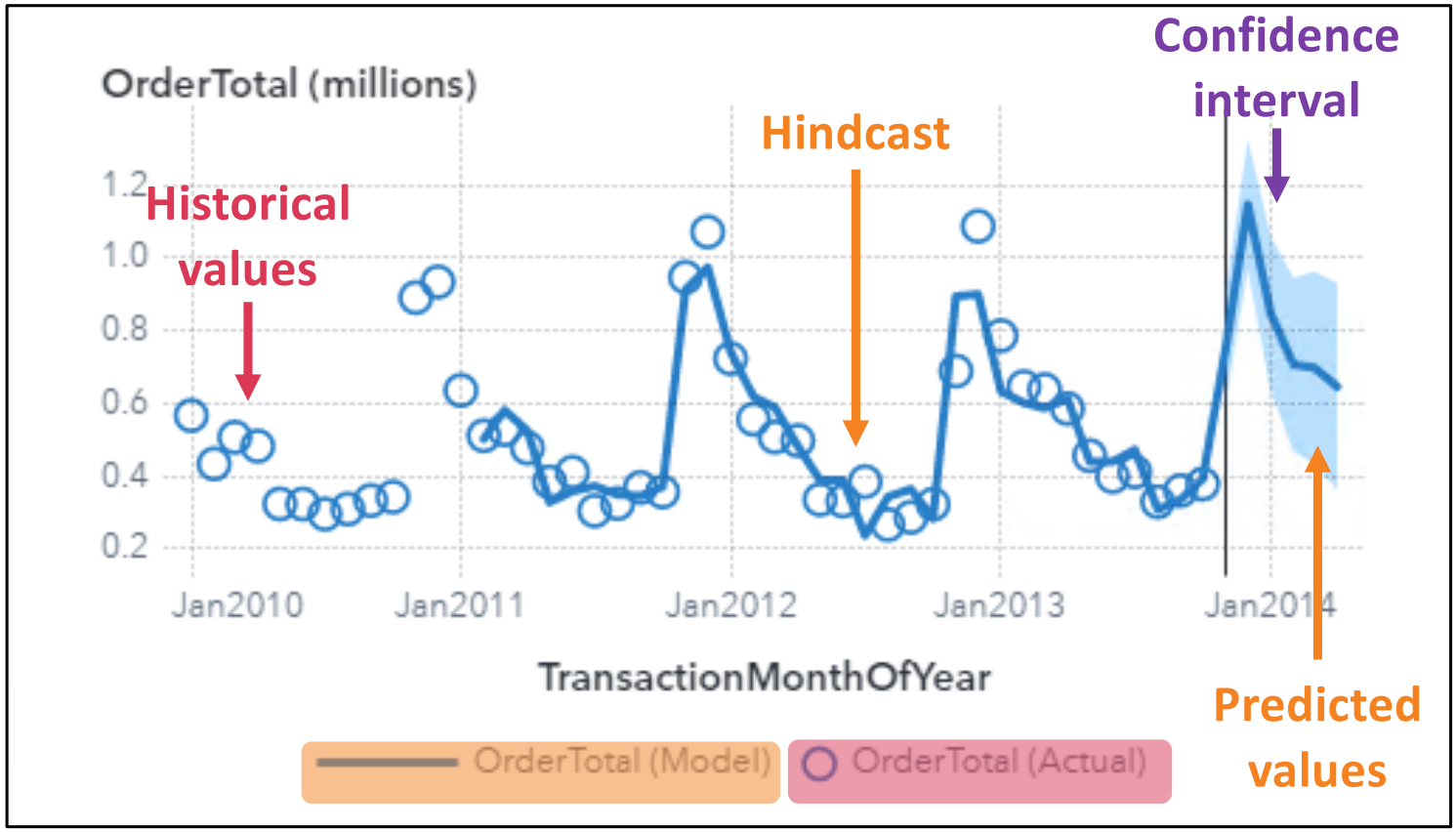
Bubble Plot with Animation Role

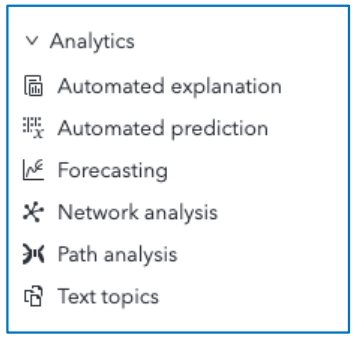




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Objects: Analytics (Forecasting)

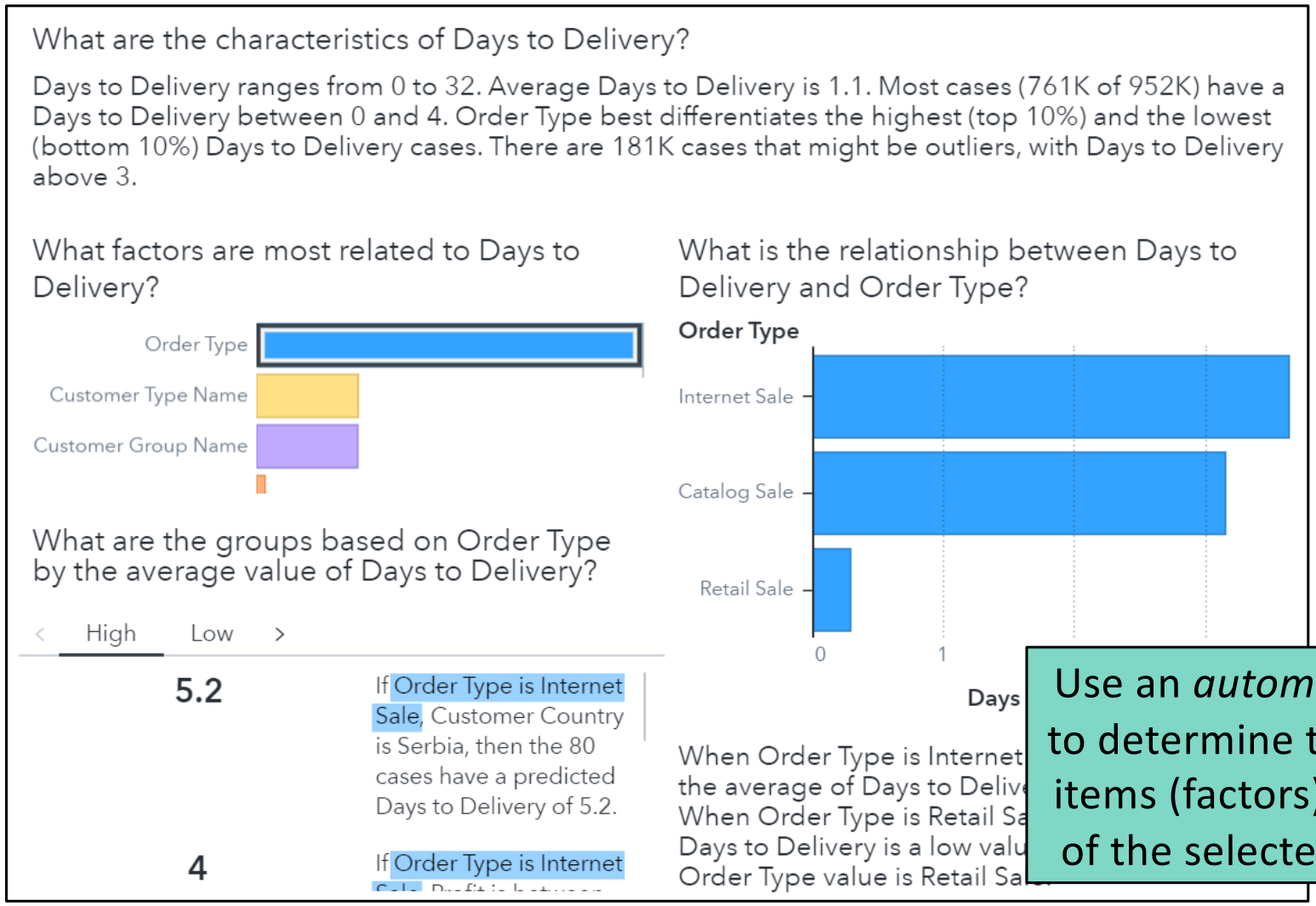




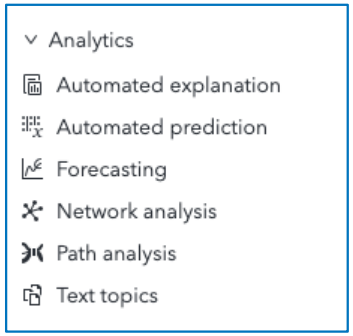
Use a *forecasting* object to show estimates of future values based on historical trends in the data.

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Objects: Analytics (Automated Explanation)



Automated explanation generates a high-level summary of the selected data item.

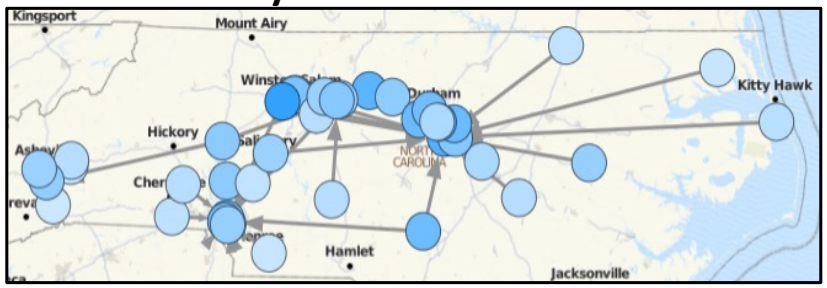


Use an *automated explanation* object to determine the most important data items (factors) that explain the values of the selected data item (response).

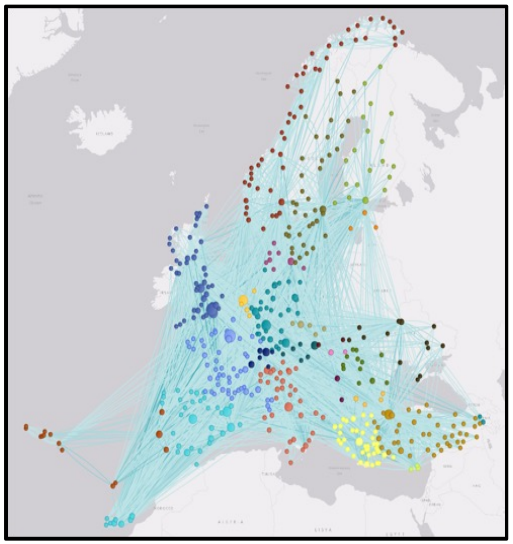
Network Analysis

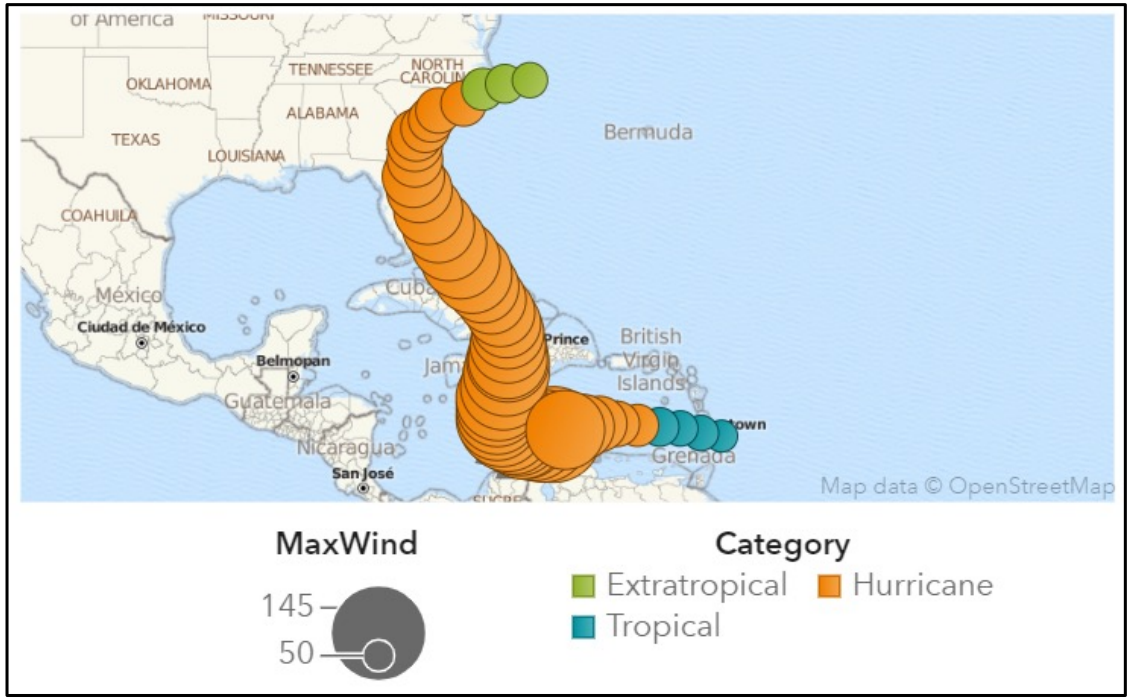
- Analytics
  - Automated explanation
  - Automated prediction
  - Forecasting
  - Network analysis
  - Path analysis
  - Text topics

Customer analysis



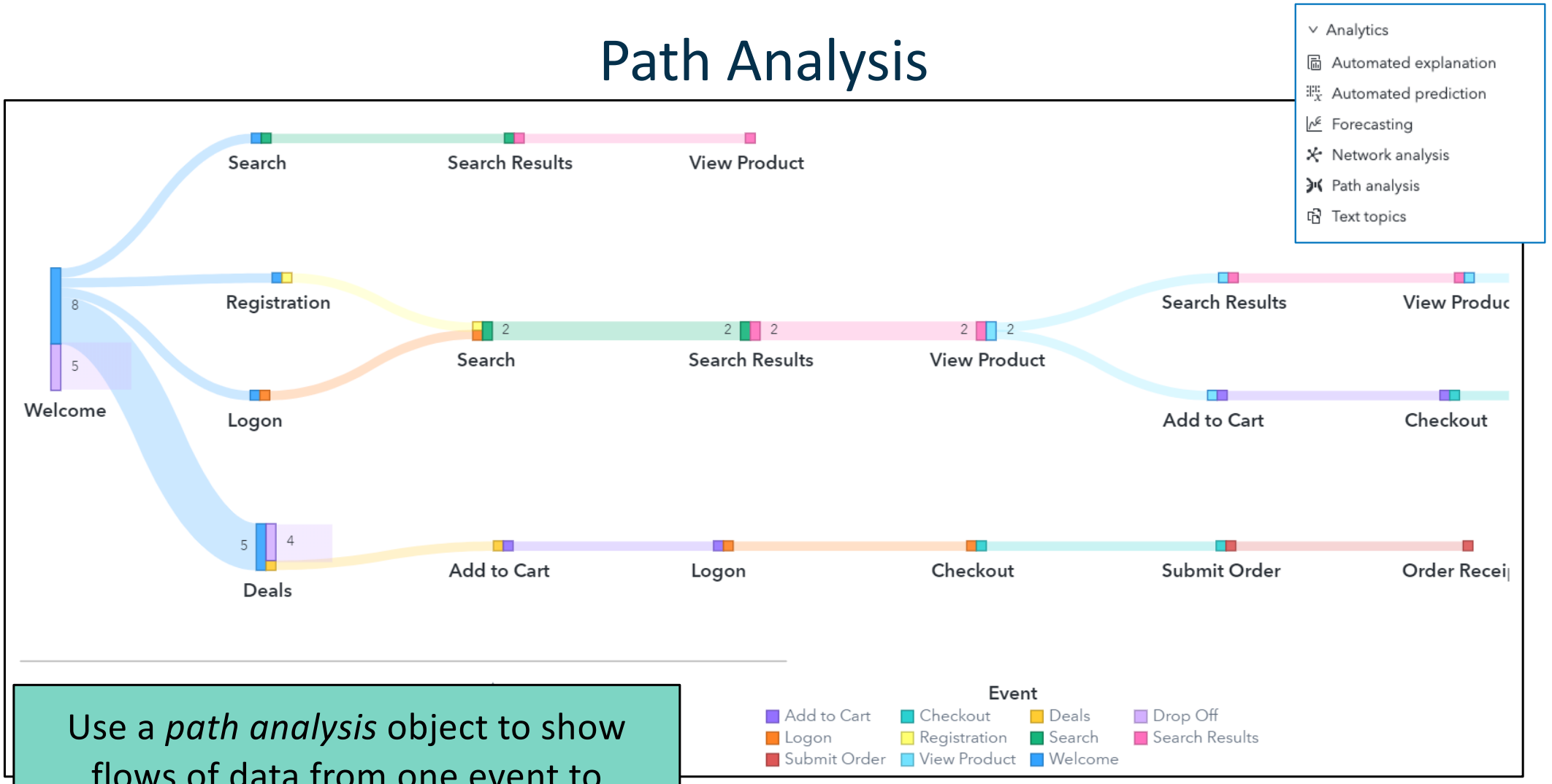
Airline connections





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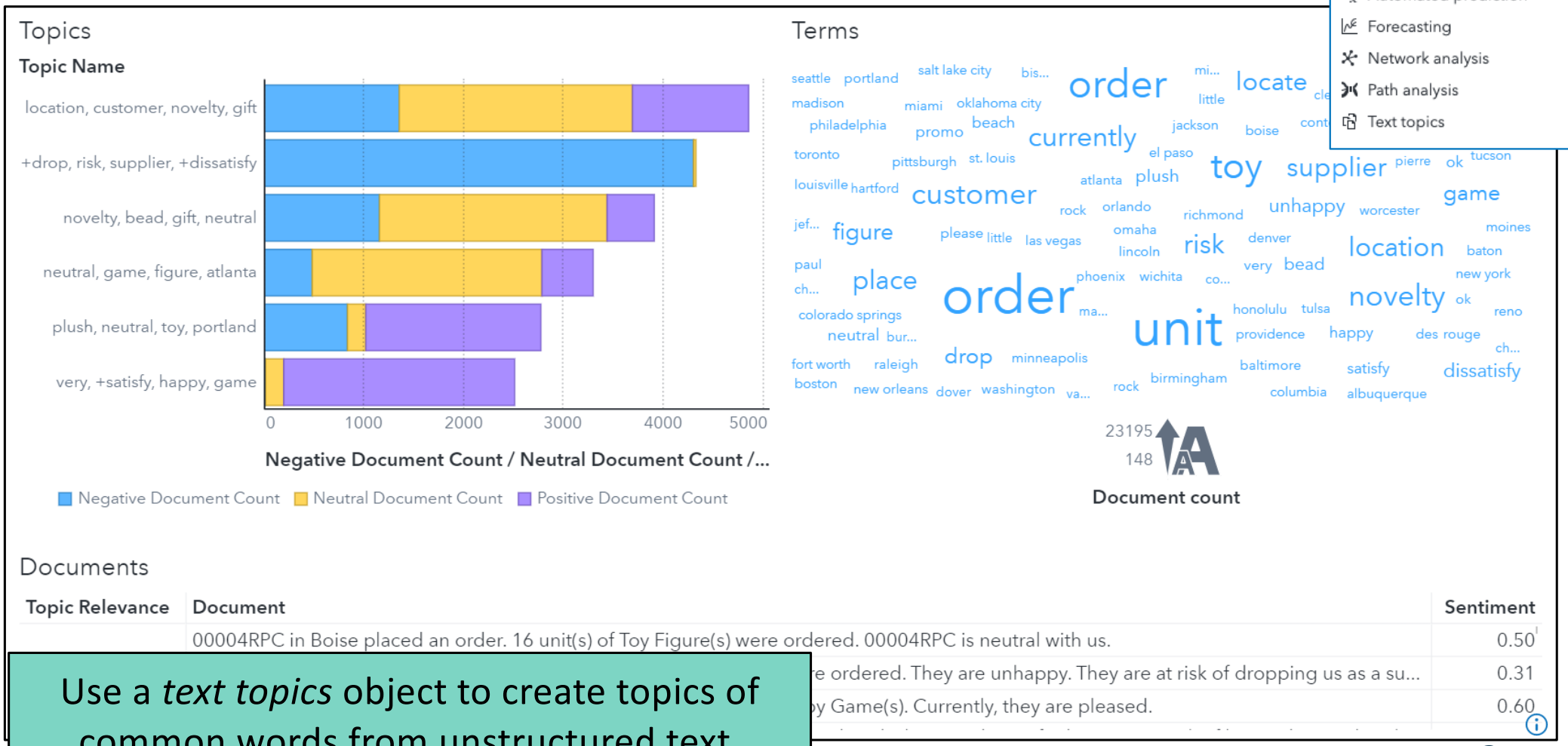
Path Analysis



Use a *path analysis* object to show flows of data from one event to another as a series of paths.

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Text topics



Documents

Topic Relevance	Document	Sentiment
	00004RPC in Boise placed an order. 16 unit(s) of Toy Figure(s) were ordered. 00004RPC is neutral with us.	0.50
	...e ordered. They are unhappy. They are at risk of dropping us as a su...	0.31
	...y Game(s). Currently, they are pleased.	0.60

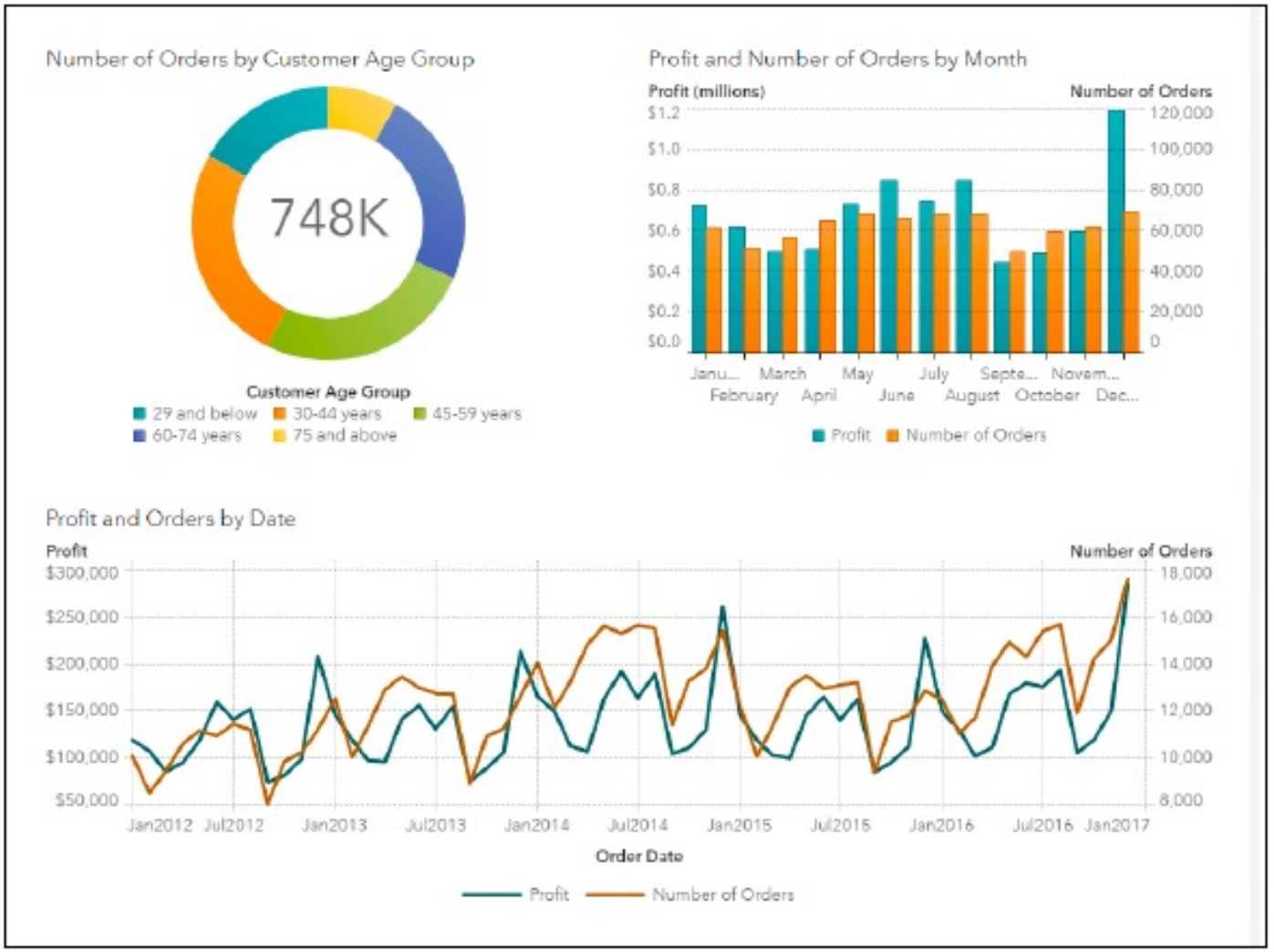
Use a *text topics* object to create topics of common words from unstructured text.

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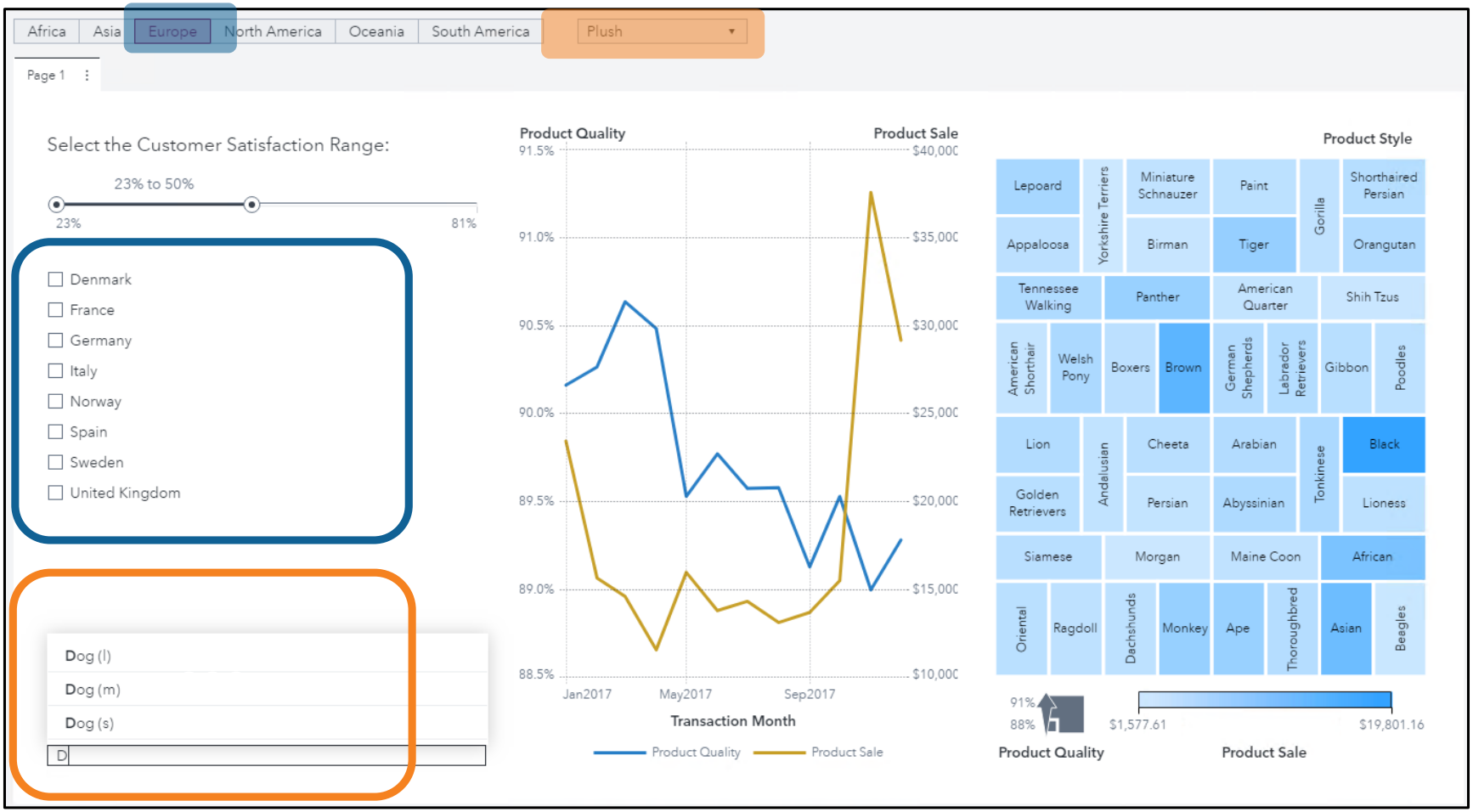
2.2) Creating interactive reports

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Dashboard


Number of Orders by Customer Age Group
748K
Customer Age Group
29 and below
30-44 years
45-59 years
60-74 years
75 and above
Profit and Number of Orders by Month
Profit (millions)
Number of Orders
$1.2
$1.0
$0.8
$0.6
$0.4
$0.2
$0.0
120,000
100,000
80,000
60,000
40,000
20,000
0
Janu...
February
March
April
May
June
July
August
Septe...
October
Novem...
Dec...
Profit
Number of Orders
Profit and Orders by Date
Profit
Number of Orders
$300,000
$250,000
$200,000
$150,000
$100,000
$50,000
18,000
16,000
14,000
12,000
10,000
8,000
Jan2012
Jul2012
Jan2013
Jul2013
Jan2014
Jul2014
Jan2015
Jul2015
Jan2016
Jul2016
Jan2017
Order Date
Profit
Number of Orders

Report and Page Prompts



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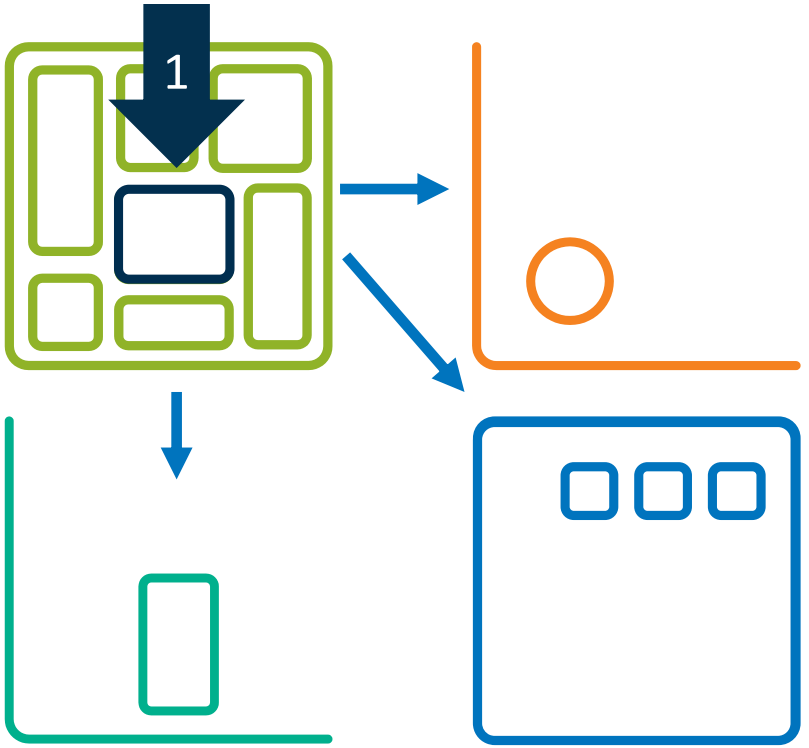
Actions



Linked selection



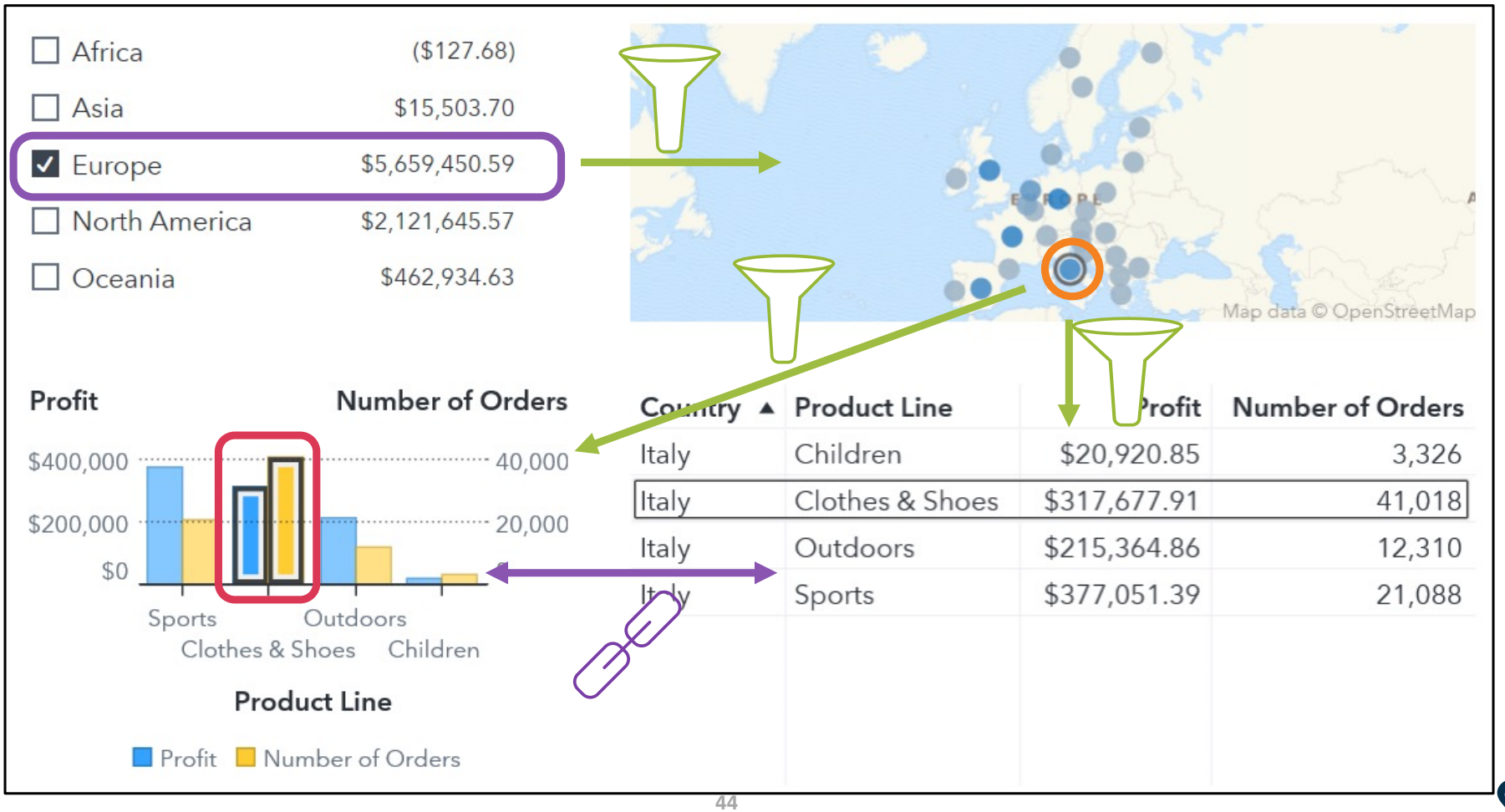
Filter



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Adding Actions to a Page



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2.3) Working with display rules

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Graph-Level Display Rules

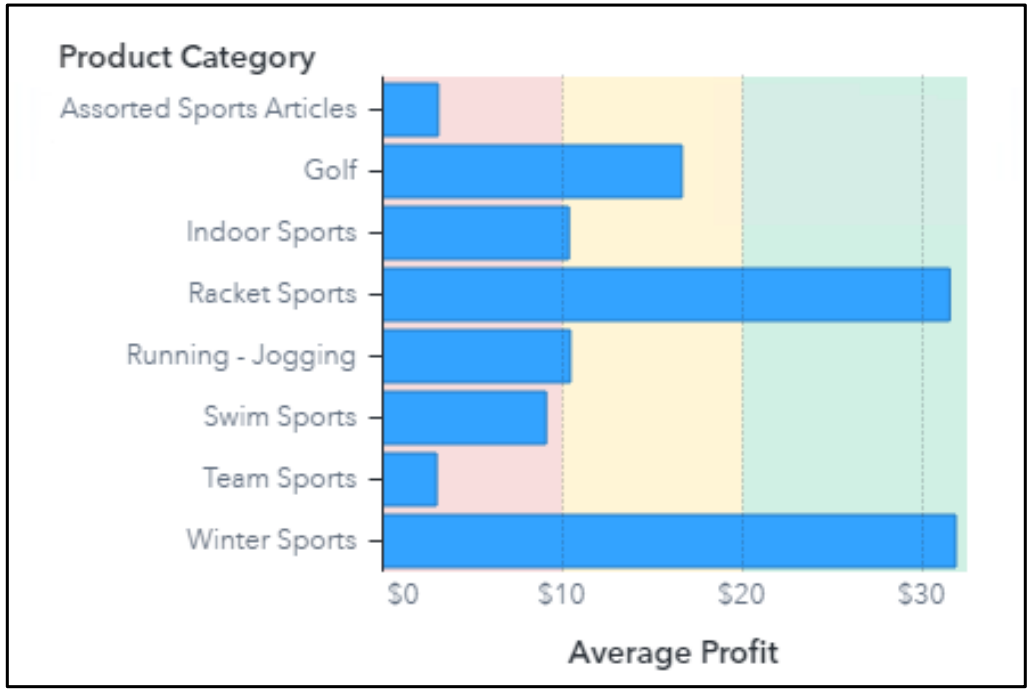
Report

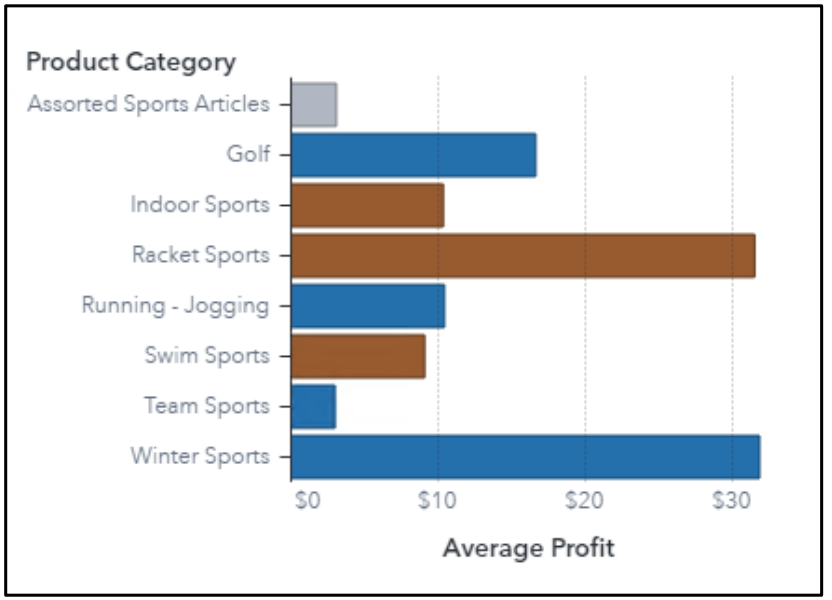
Expression

123

Color-mapped values

ABC





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Table-Level Display Rules

Report

Color-mapped values



Product Line ▲	Order Type	Profit	Number of Orders
Outdoors	Catalog Sale	$240,285	17,405
Outdoors	Internet Sale	$219,350	14,746
Outdoors	Retail Sale	$1,227,450	75,465
Sports	Catalog Sale	$623,088	44,946
Sports	Internet Sale	$518,276	38,605
Sports	Retail Sale	$3,049,267	221,875

Expression



Gauge



Product Line ▲	Order Type	Profit	Number of Orders
Outdoors	Catalog Sale	$240,285	17,405
Outdoors	Internet Sale	$219,350	14,746
Outdoors	Retail Sale	$1,227,450	75,465
Sports	Catalog Sale	$623,088	44,946
Sports	Internet Sale	$518,276	38,605
Sports	Retail Sale	$3,049,267	221,875

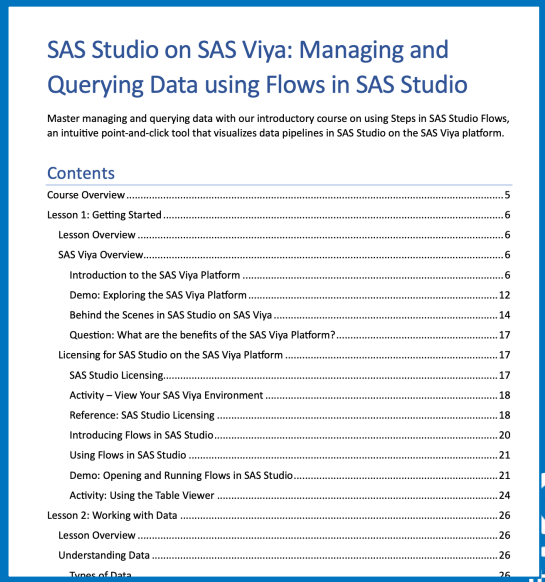
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3) SAS Studio

Data Engineering and Preparation with SAS

What is SAS Studio? + SAS Studio Flow Lab



SAS Studio on SAS Viya: Managing and Querying Data using Flows in SAS Studio

Master managing and querying data with our introductory course on using Steps in SAS Studio Flows, an intuitive point-and-click tool that visualizes data pipelines in SAS Studio on the SAS Viya platform.

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Lesson Overview 6
SAS Viya Overview 6
Introduction to the SAS Viya Platform 6
Demo: Exploring the SAS Viya Platform 12
Behind the Scenes in SAS Studio on SAS Viya 14
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Licensing for SAS Studio on the SAS Viya Platform 17
SAS Studio Licensing 17
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Understanding Data 26
Types of Data 26

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What Is SAS Studio?



point-and-click

Program Editor

flows

tasks

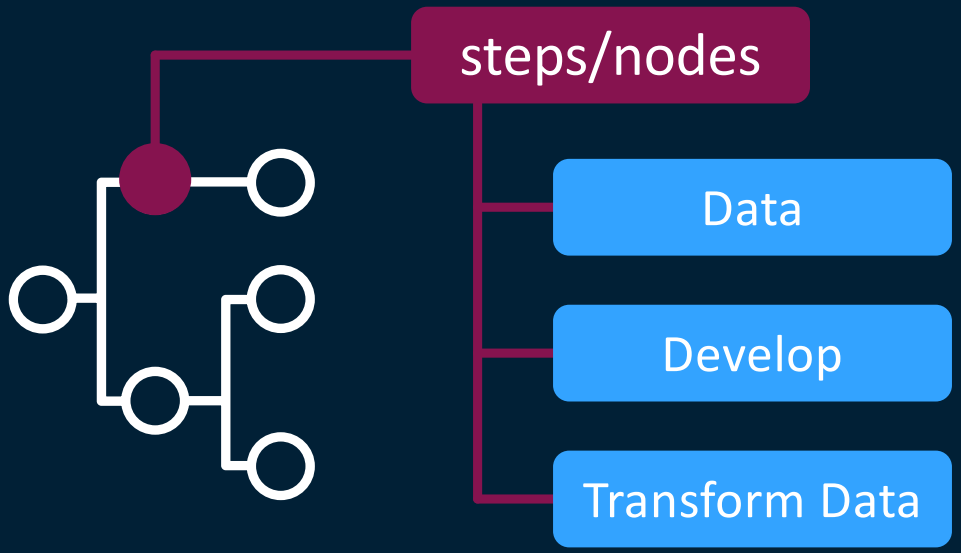
snippets

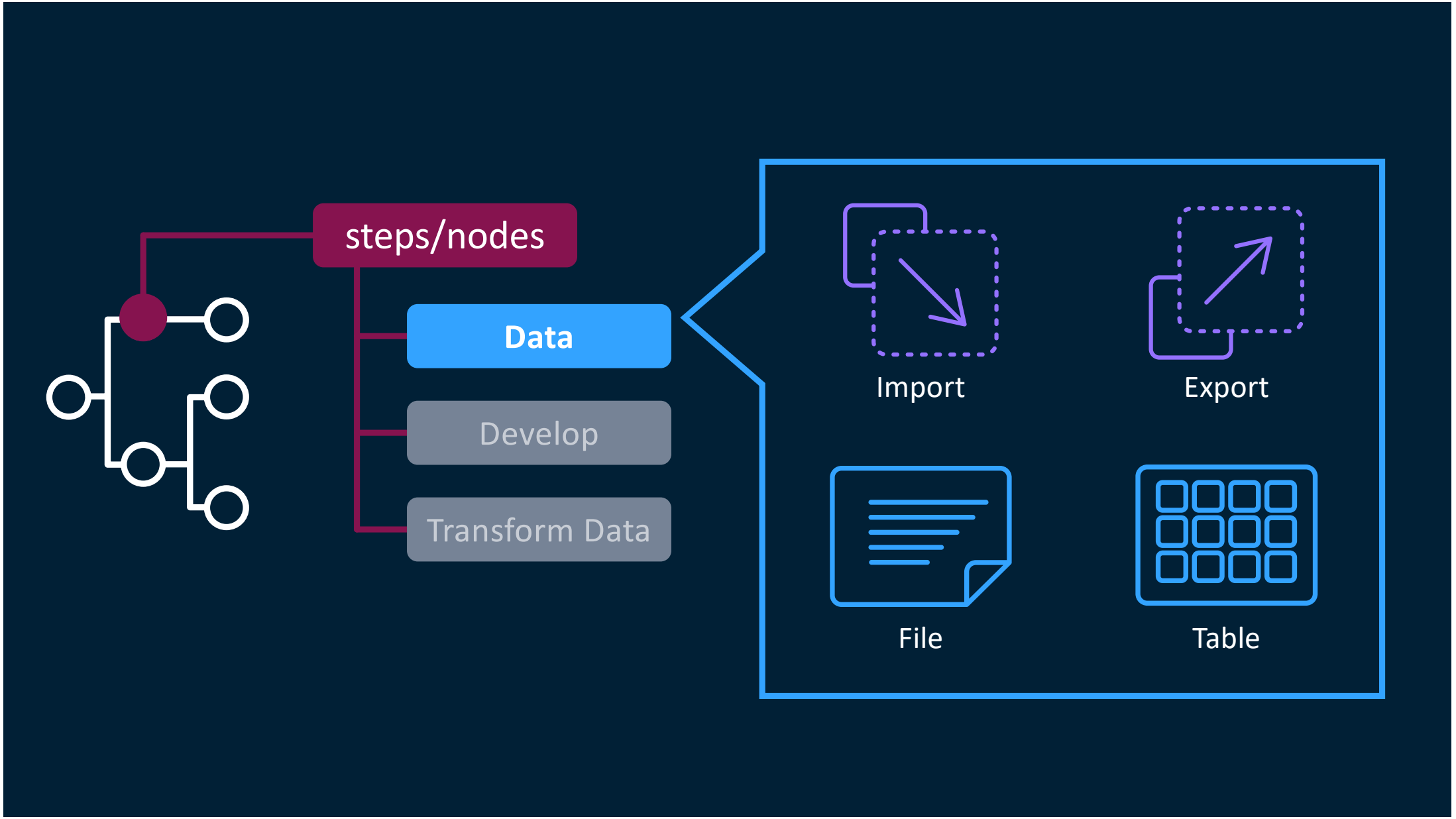
syntax Help
and autocomplete

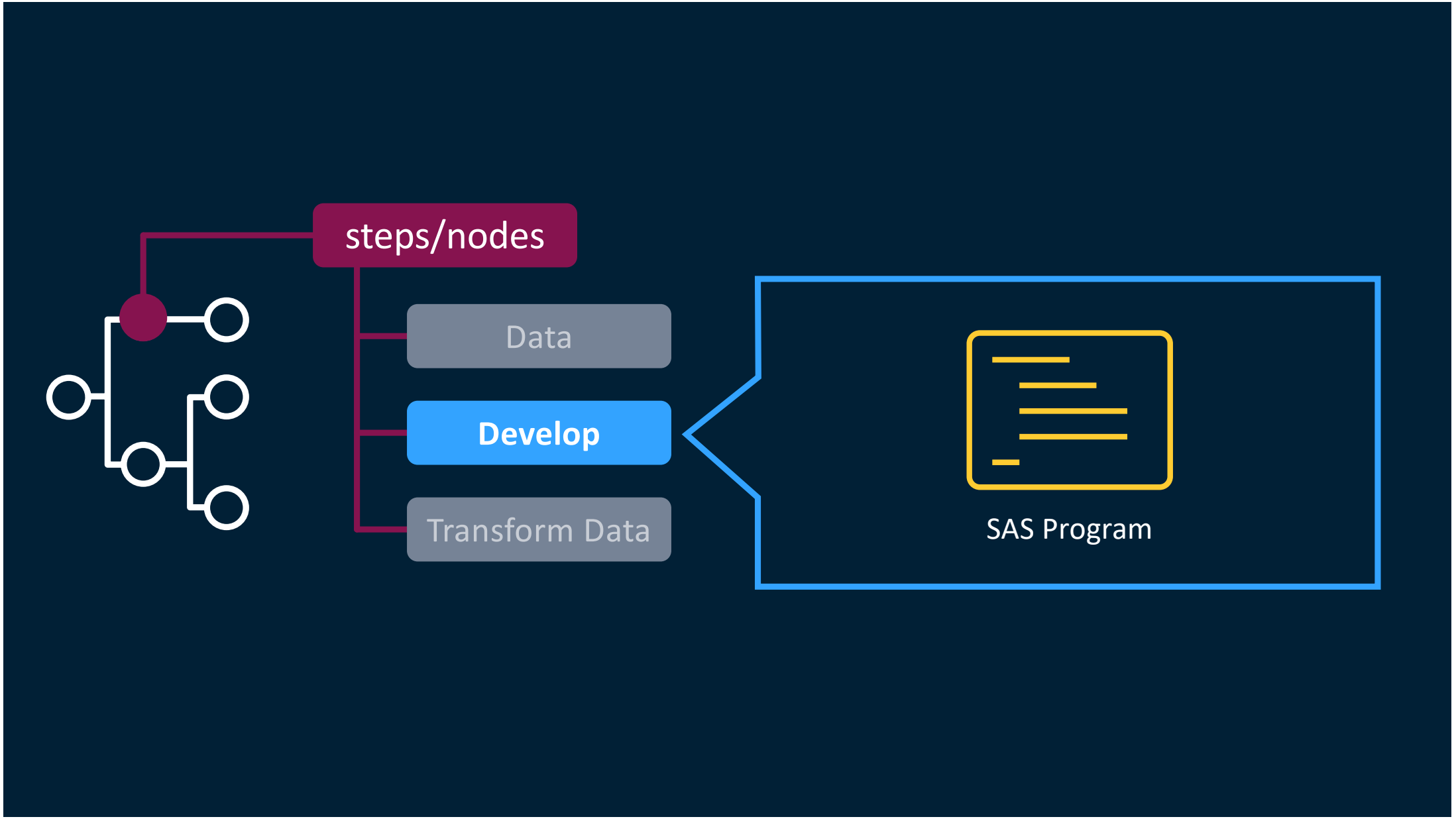
da

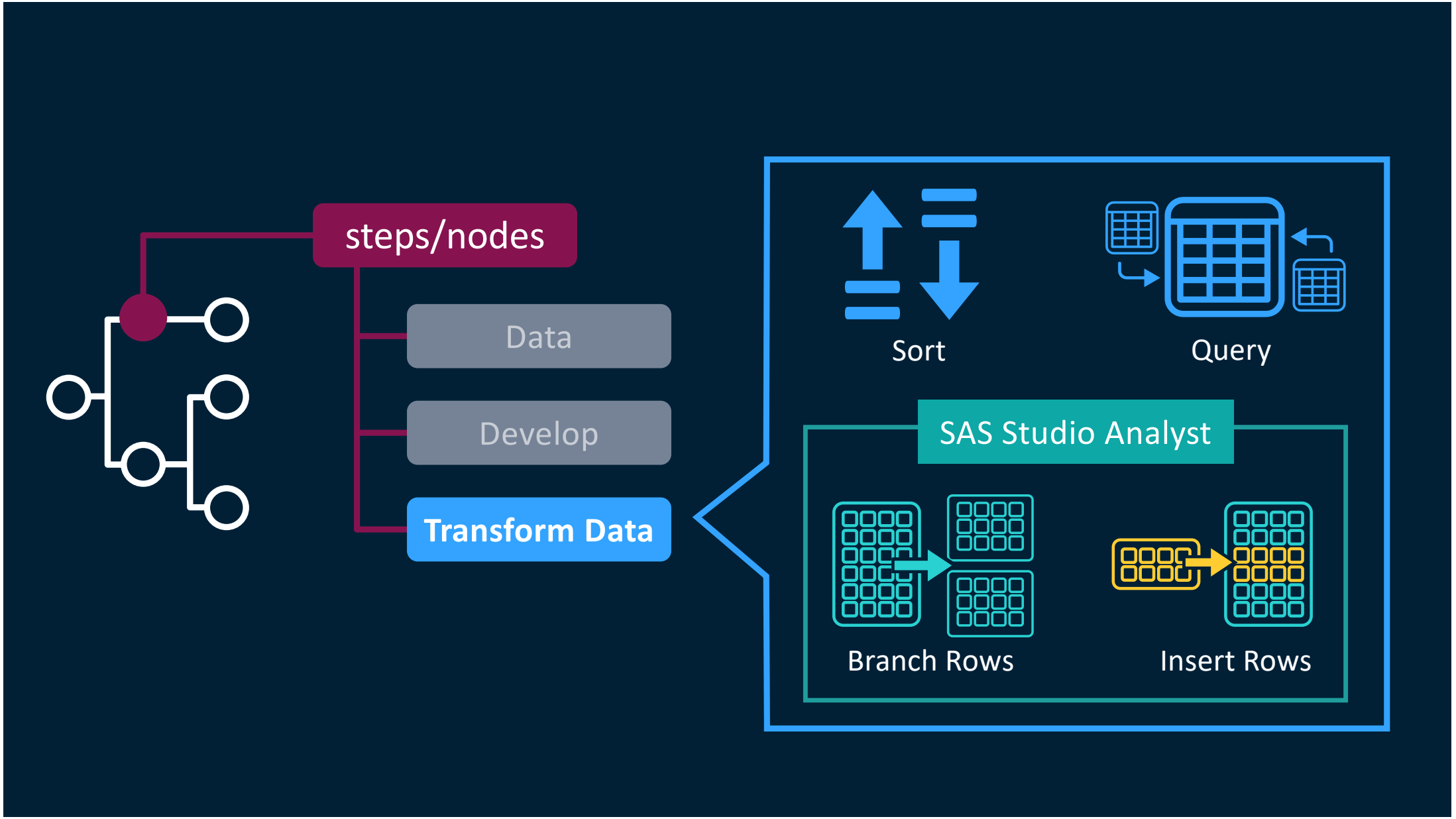
data

SAS Studio Flows

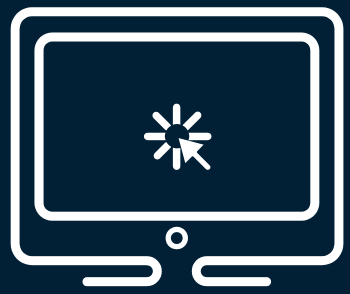



steps/nodes
Data
Develop
Transform Data
Import
Export
File
Table

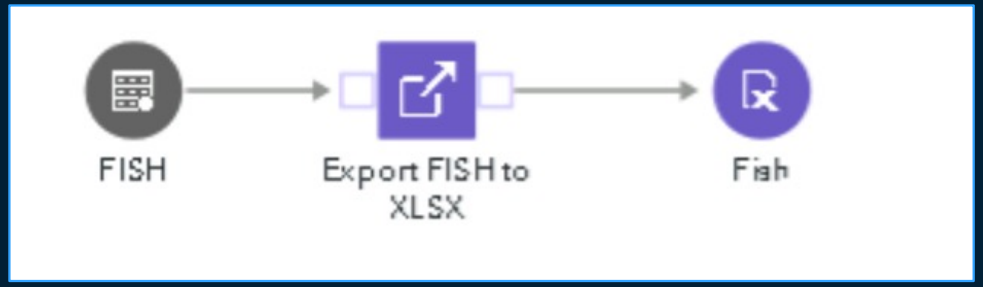

steps/nodes
Data
Develop
Transform Data
SAS Program

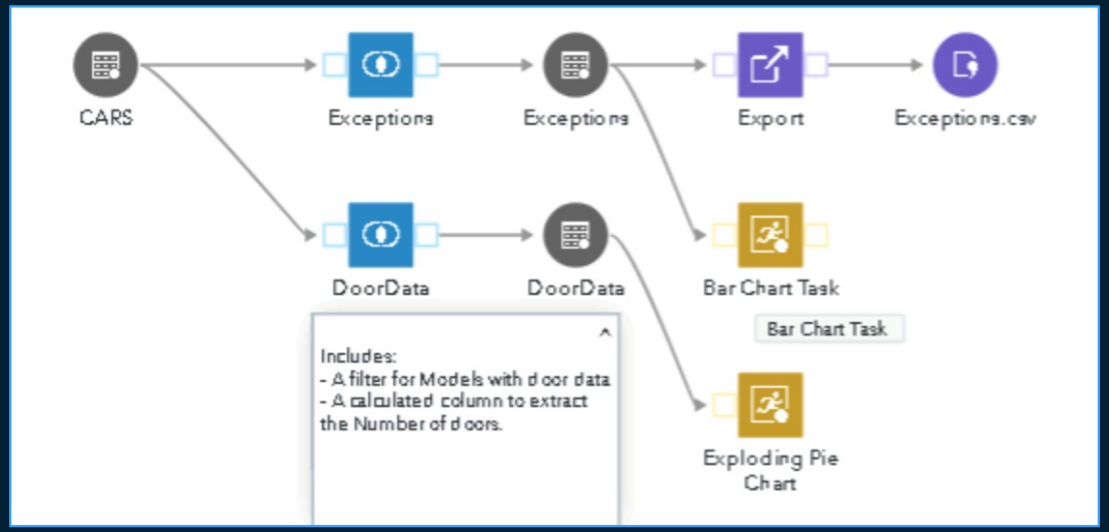

steps/nodes
Data
Develop
Transform Data
Sort
Query
SAS Studio Analyst
Branch Rows
Insert Rows

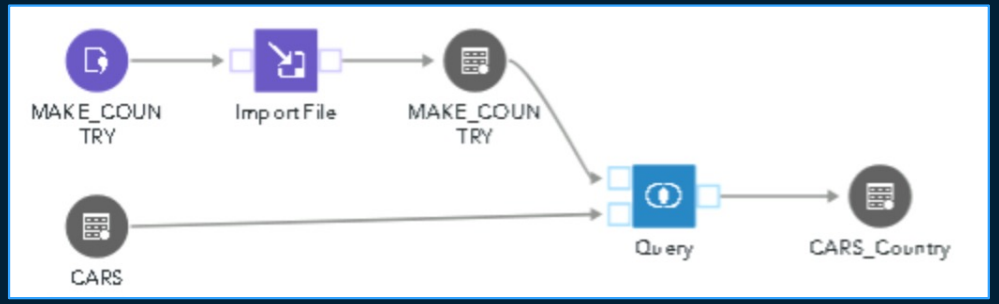
Lab: SAS Studio Flow













5) SAS Model Studio

Data Engineering and Preparation with SAS

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1) Machine Learning

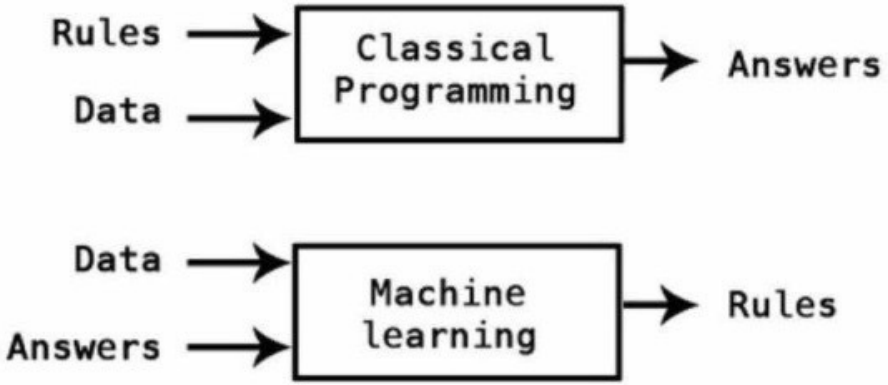
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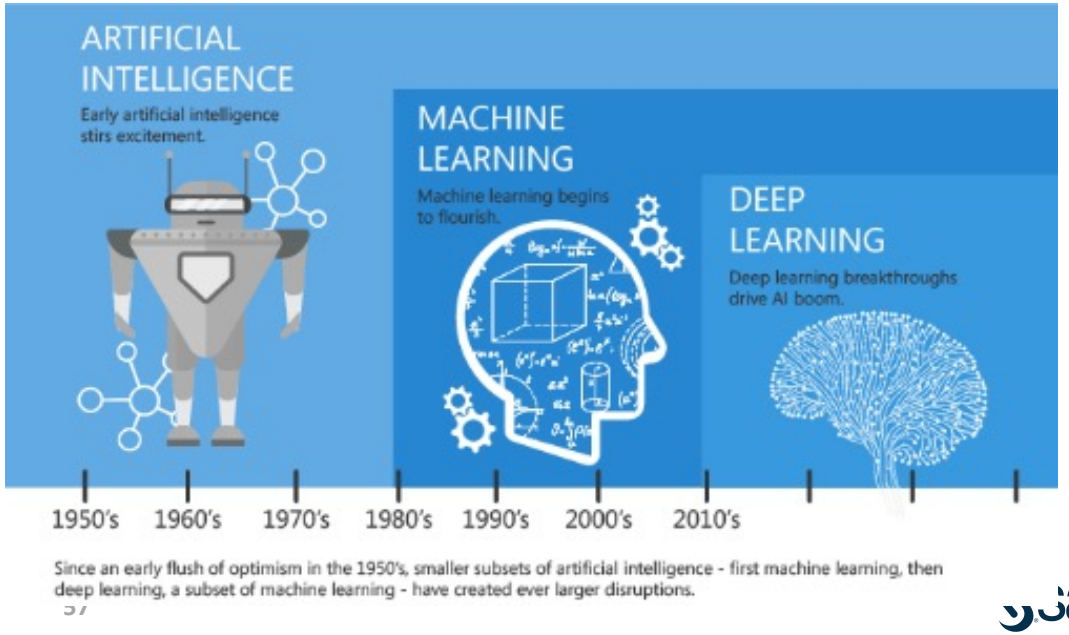
AI, Machine Learning, and Deep Learning

Machine Learning (ML) is a subfield in AI focusing on making to learn by itself without human intervention.

"Machine learning is the sc*ience of getting computers to act without being explicitly programmed*." — Stanford University

https://towardsdatascience.com/cousins-of-artificial-intelligence-dda4edc27b55





Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

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Terminology: Data table

	inputs			target
Age	**Income**	**Gender**	**Province**	**Default**
25	25,000	Female	Bangkok	Bad
35	50,000	Female	Nontaburi	Good
32	35,000	Male	Bangkok	Bad

Row

- Example, instance, case, observation, subject

Column

- Feature, variable, attribute

Input

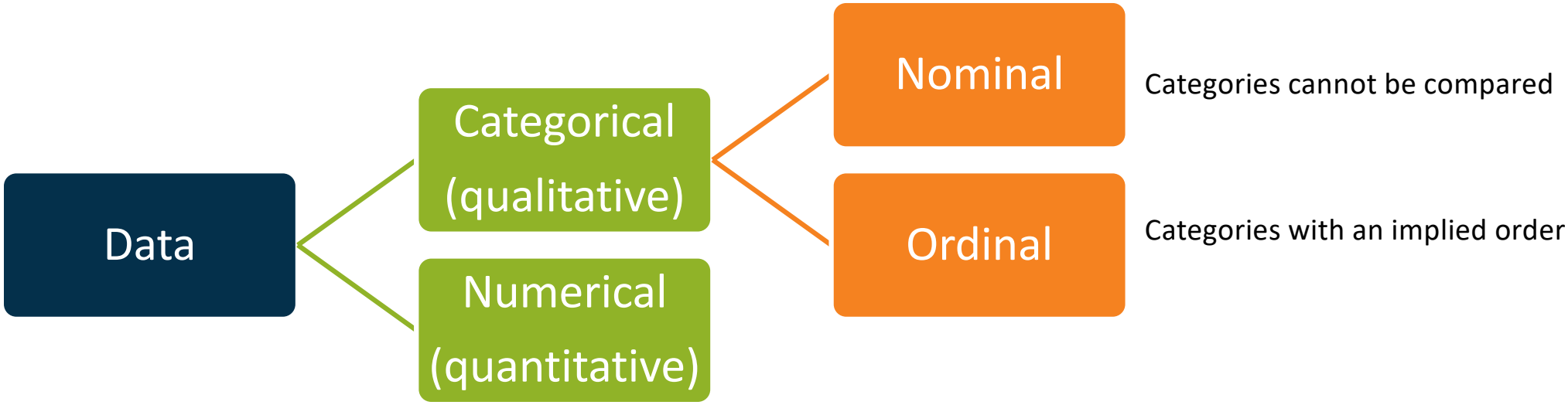
- Predictor, independent, explanatory variable

Target

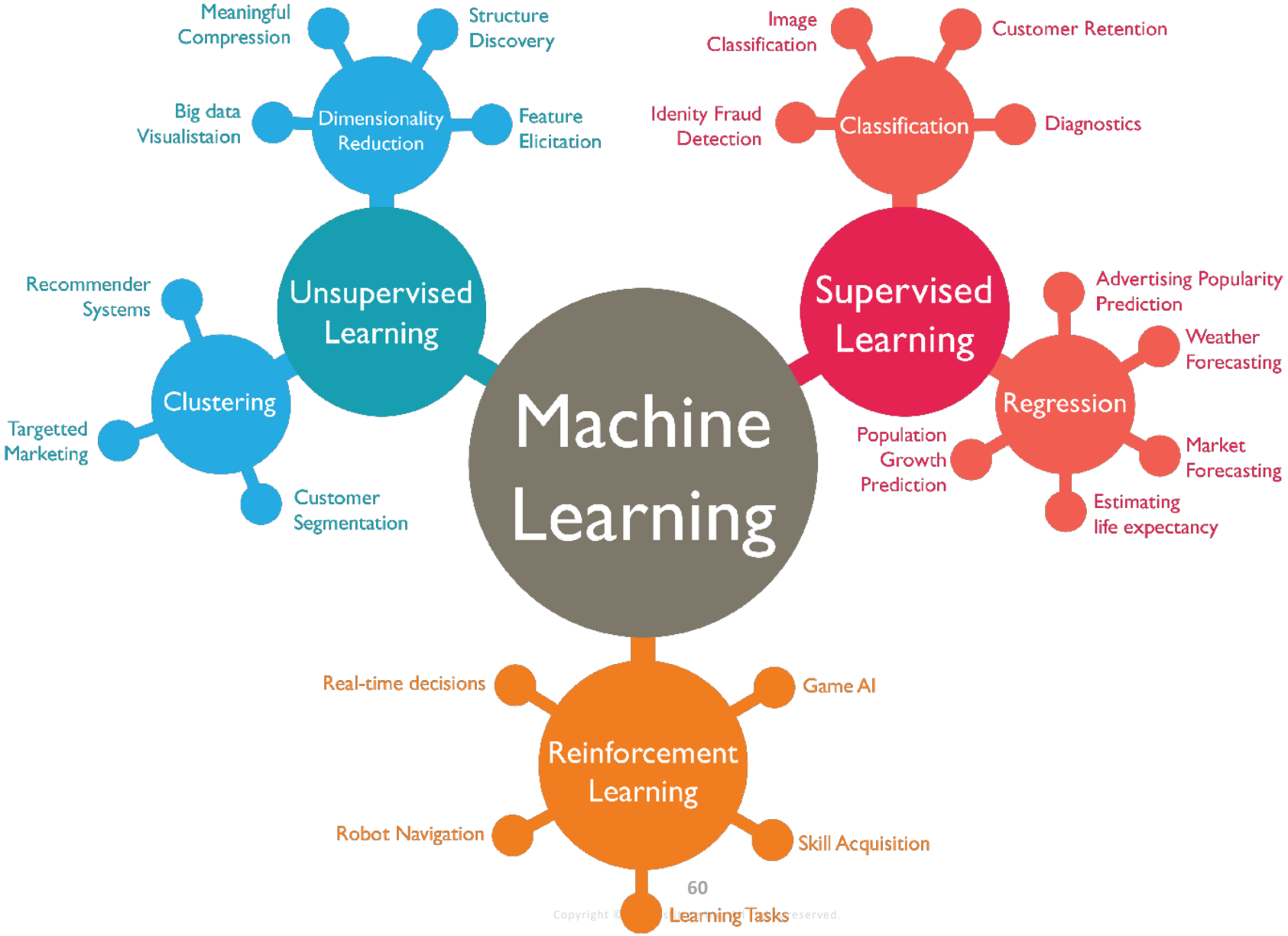
- Output, outcome, response, dependent variable

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Terminology: Kinds of data



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Task1: Supervised learning

Handcrafted features

Training Data



inputs | target

Age	Income	Gender	Province	Default
25	25,000	Female	Bangkok	Bad
35	50,000	Female	Nontaburi	Good
32	35,000	Male	Bangkok	Bad

Testing Data



Age	Income	Gender	Province	Default
25	25,000	Female	Bangkok	?

Application: Direct Target Marketing

Prediction algorithms

Decision Tree

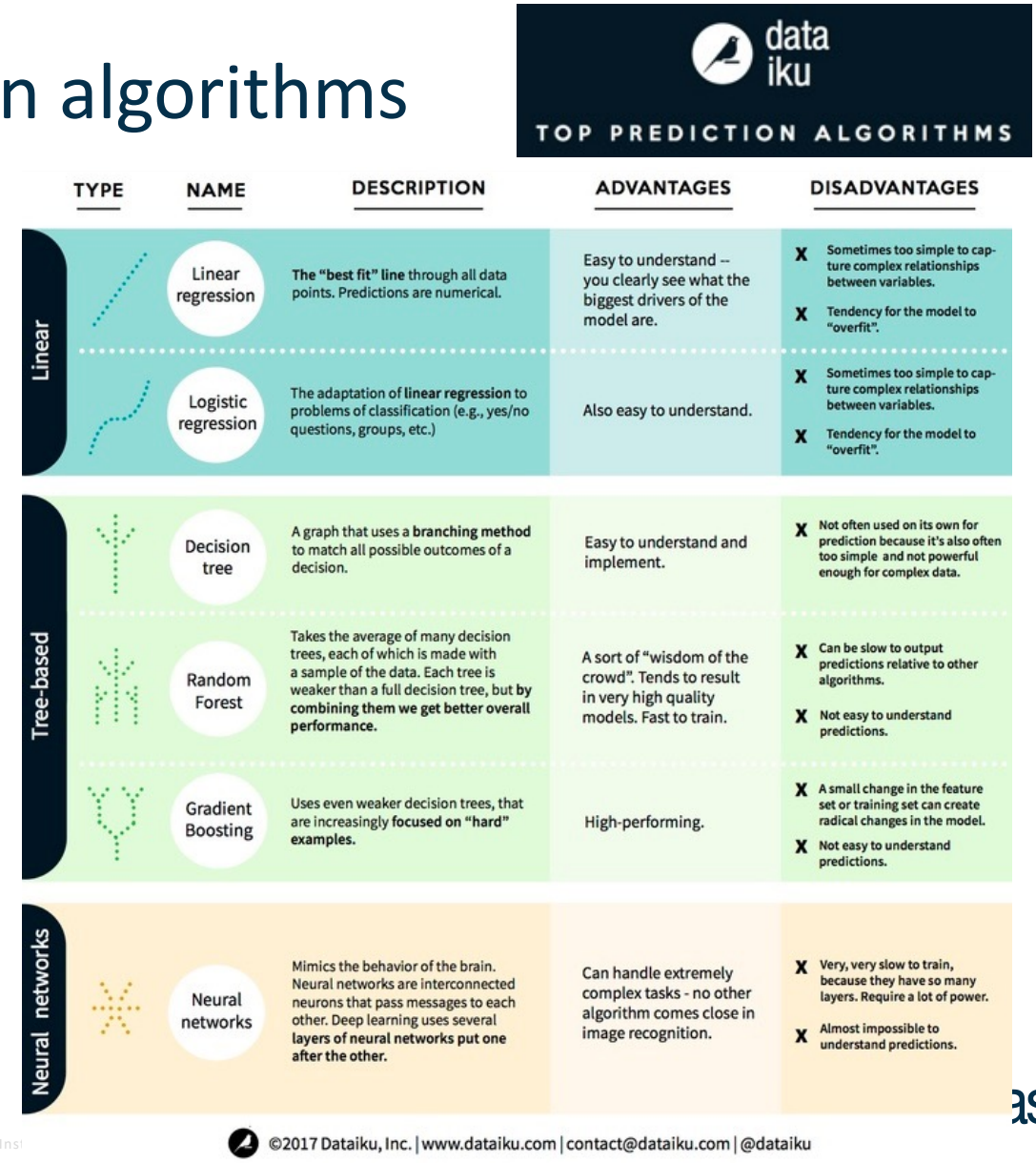
(Logistic) Regression

kNN

Support Vector Machine

Neural Networks (NN)

Deep Learning



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Task2: Unsupervised learning (descriptive task)

Training Data



inputs

target

Age	Income	Gender	Province	chase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	

Testing Data



Age	Income		Province	Purchase
25	25,000		Bangkok	?



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Clustering



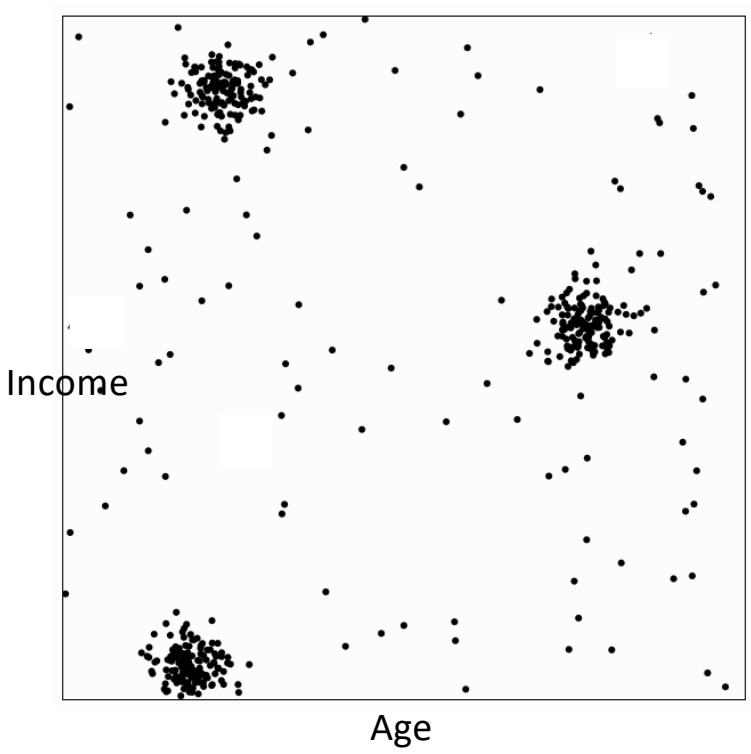


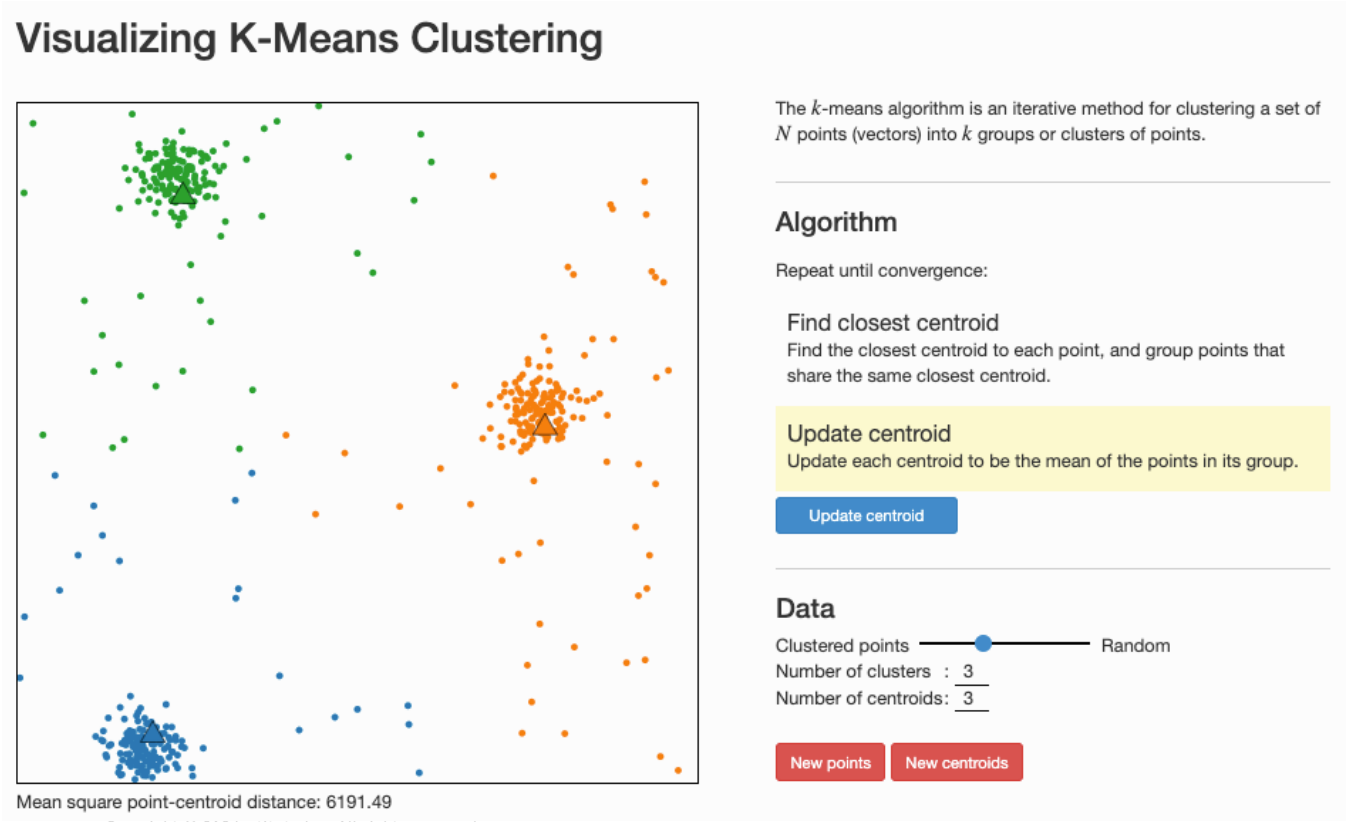
- In our class, there are many participants. Should we teach them using the same method?
- May be not! Since they may have different learning behaviors and backgrounds.
- Inputs
  - Education field
  - Level of English communication
  - Level of computer skills
  - Age range
  - Gender

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K-means Clustering

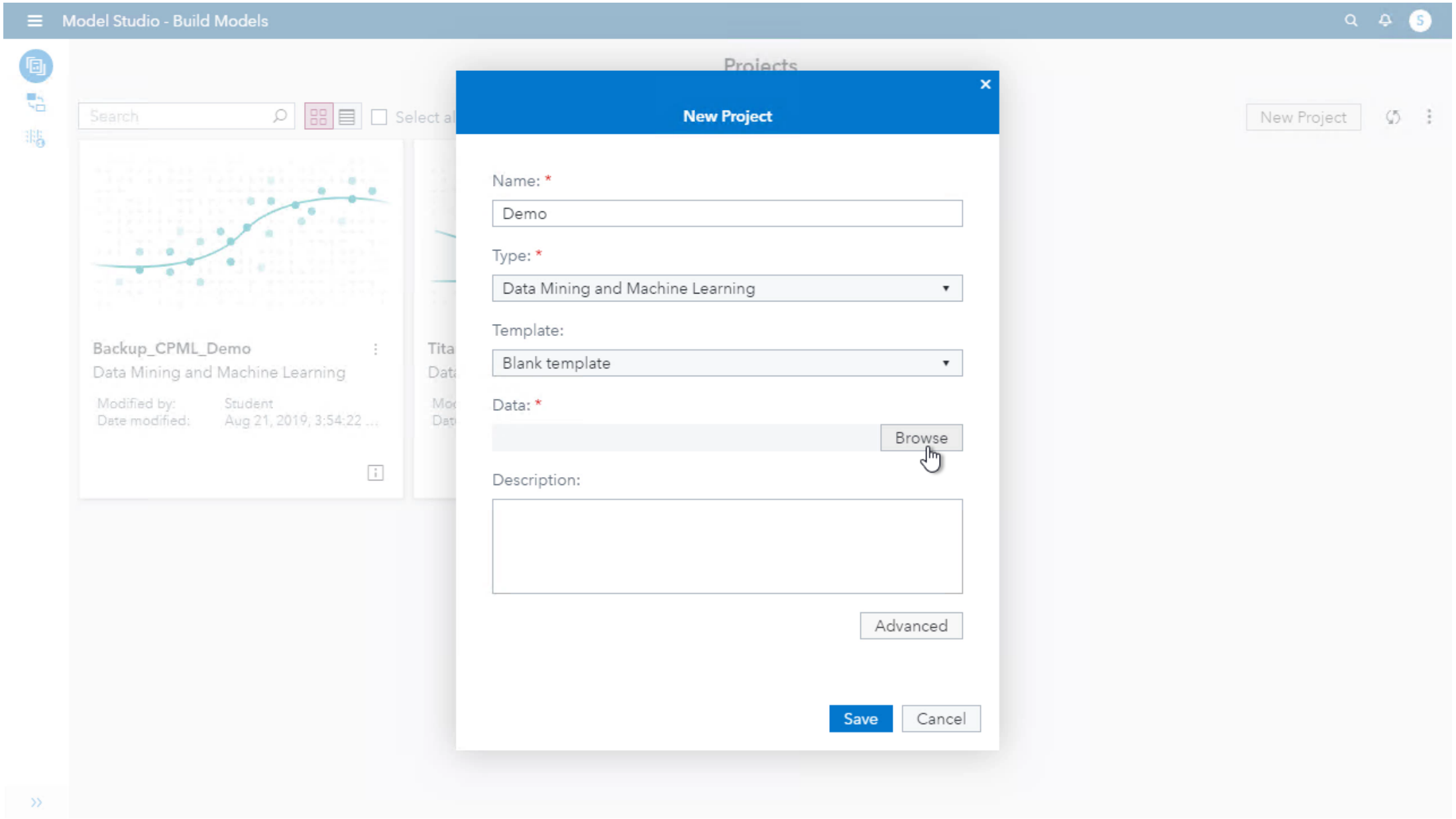
http://web.stanford.edu/class/ee103/visualizations/kmeans/kmeans.html





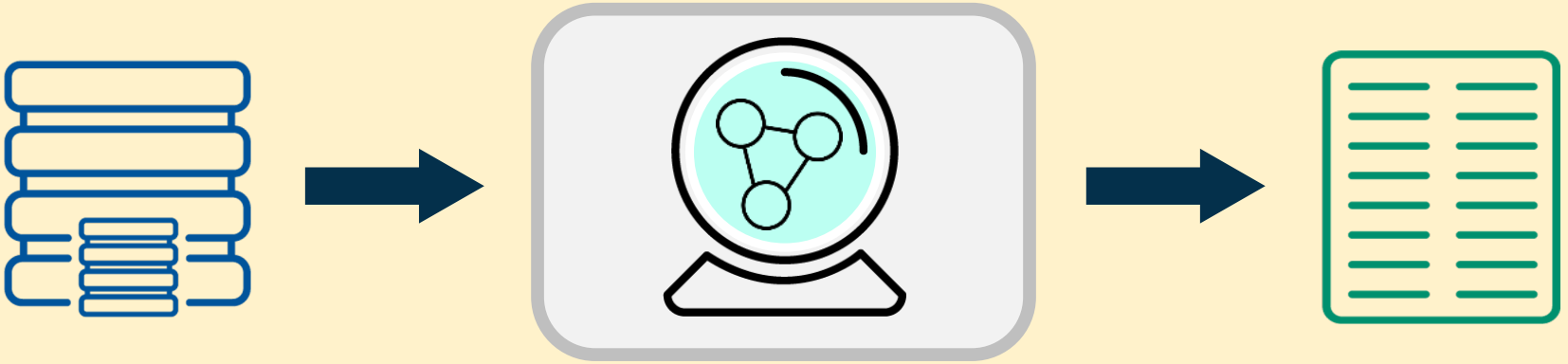
2) Introduction to SAS Model Studio

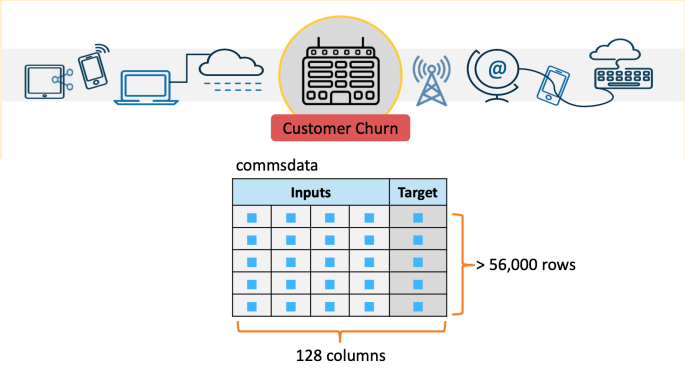
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Model Studio - Build Models
Projects
Search
Select all
New Project
Backup_CPML_Demo
Data Mining and Machine Learning
Modified by: Student
Date modified: Aug 21, 2019, 3:54:22 ...
New Project
Name: *
Demo
Type: *
Data Mining and Machine Learning
Template:
Blank template
Data: *
Browse
Description:
Advanced
Save
Cancel

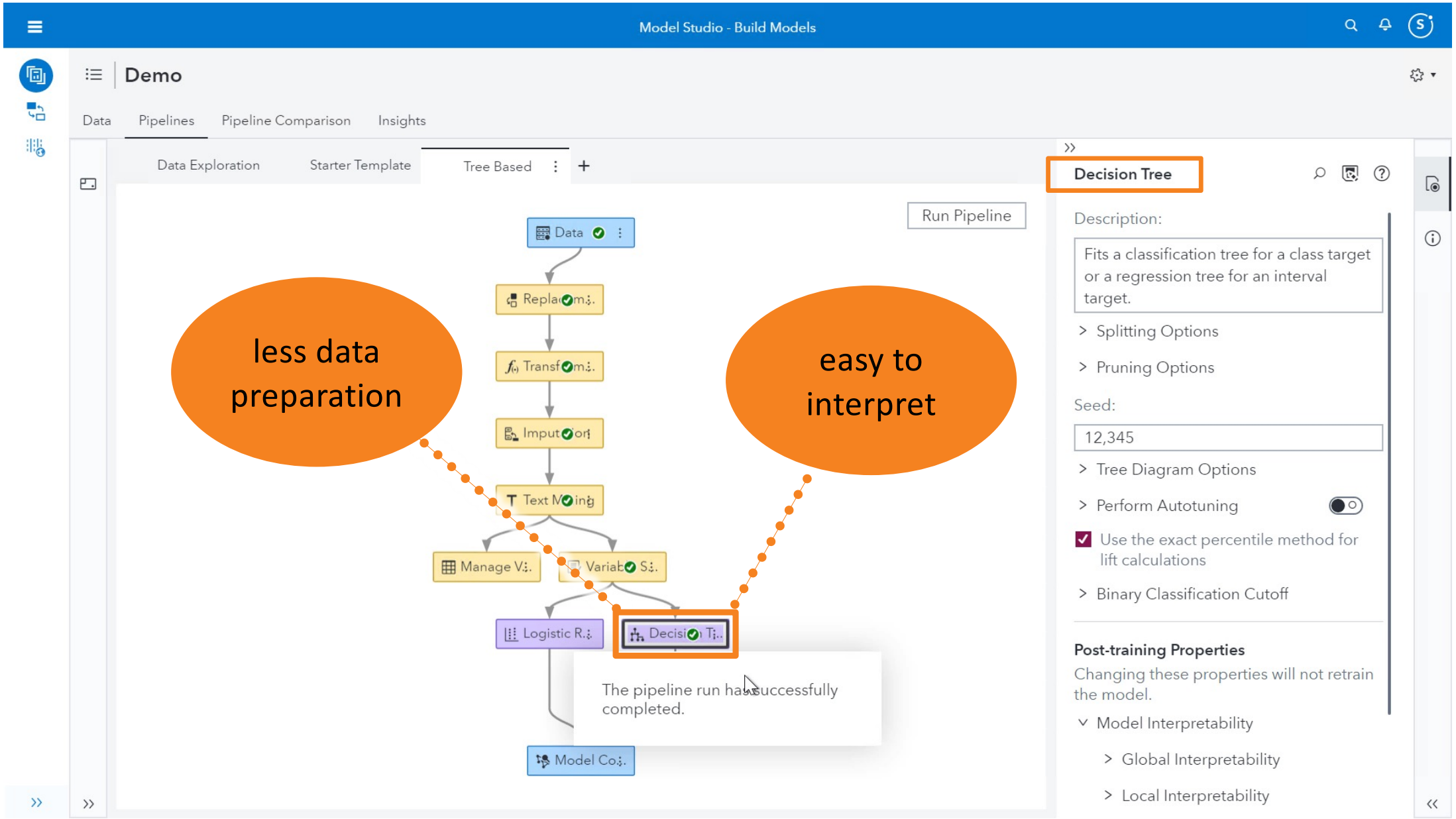
Machine Learning

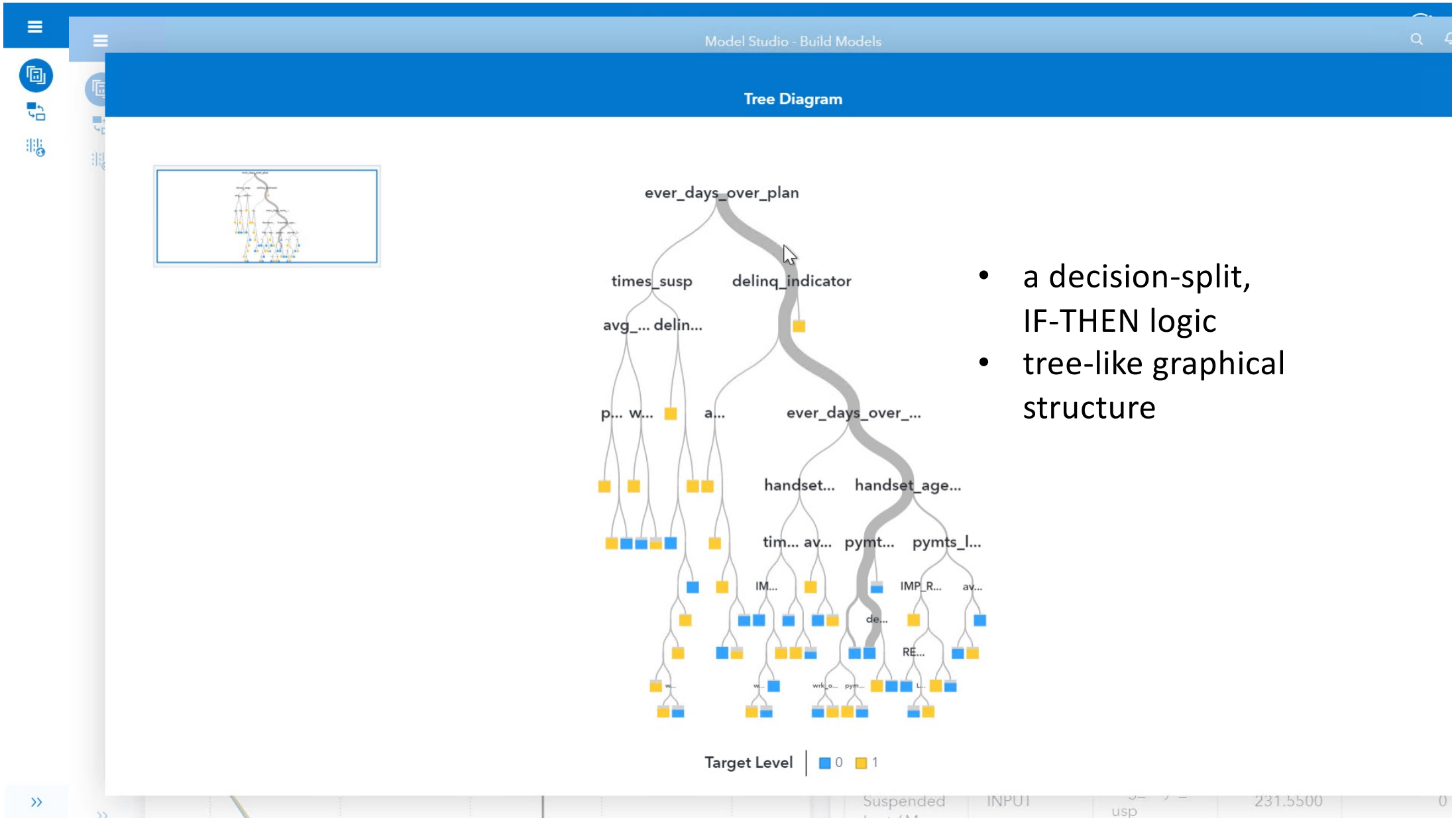
Automation | Customization | Acceleration





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Model Studio - Build Models
Demo
Data
Pipelines
Pipeline Comparison
Insights
Data Exploration
Starter Template
Tree Based
Run Pipeline
less data preparation
easy to interpret
Data
Logistic R...
Model Co...
The pipeline run has successfully completed.
Decision Tree
Description:
Fits a classification tree for a class target or a regression tree for an interval target.
Splitting Options
Pruning Options
Seed:
12,345
Tree Diagram Options
Perform Autotuning
Use the exact percentile method for lift calculations
Binary Classification Cutoff
Post-training Properties
Changing these properties will not retrain the model.
Model Interpretability
Global Interpretability
Local Interpretability



- a decision-split, IF-THEN logic
- tree-like graphical structure

6) Data Governance with SAS

Lab only & no slides

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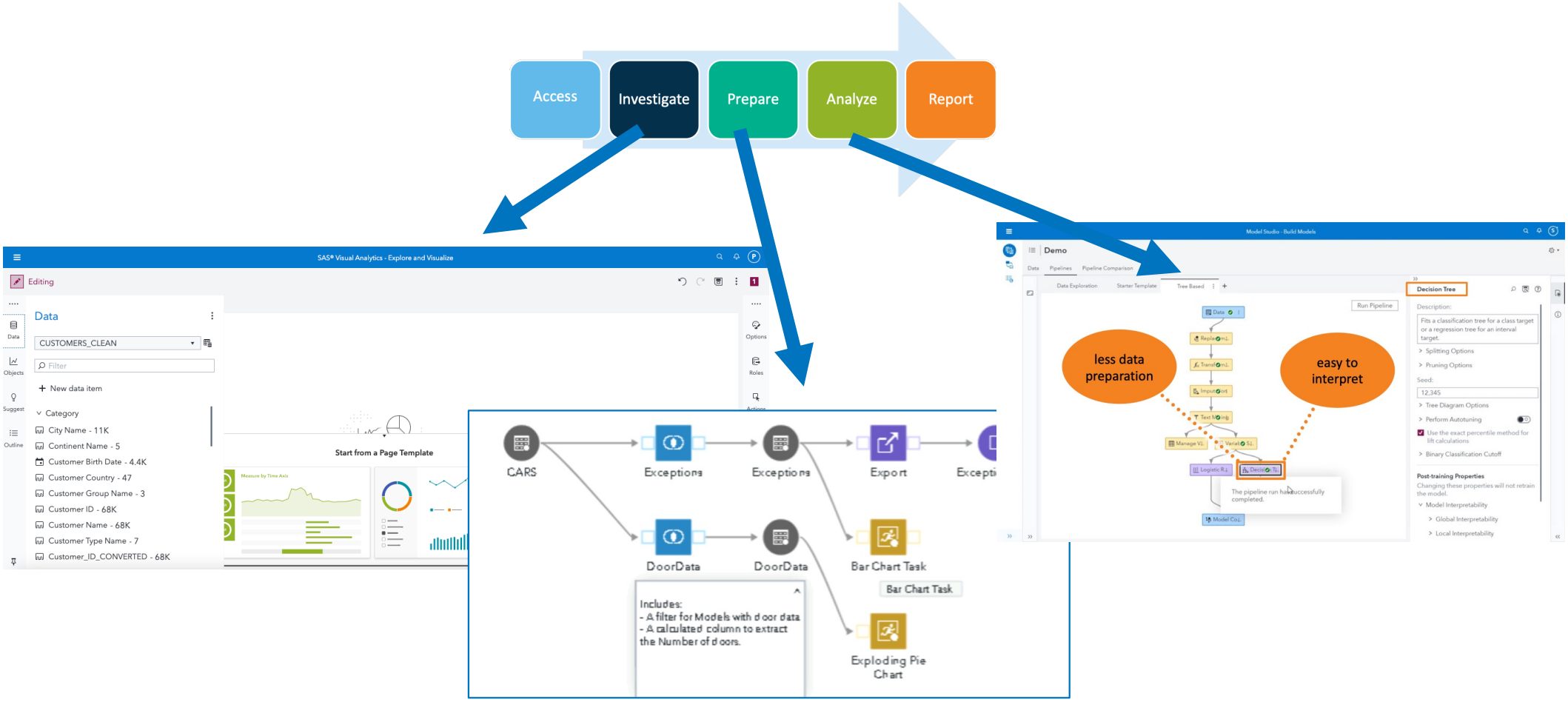
7) Conclusion



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The Analytics Journey Overview



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Thank you very much & any questions

peerapon.v@chula.ac.th

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